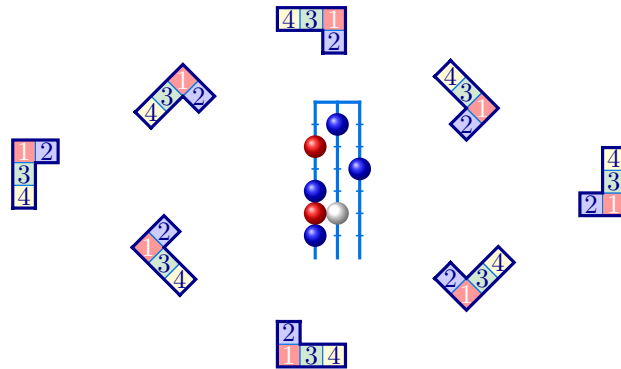


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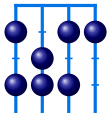


1. INTRODUCTION

The package provides commands for drawing common objects in algebraic combinatorics and representation theory, together with styling options and a key-value interface for additional customization. Under the hood, everything is drawn using `TikZ` (and `LATEX3`). These commands can be used either as standalone commands or as part of a `tikzpicture` environment.

This package grew out of my own research, so it does all of the things that I want it to do. On the other hand, it is written as flexibly as possible in the hope that it will draw the diagrams that others need.

For the impatient, here is a basic example that shows how to use the main commands provided by the `aTableau` package:



1	2	3	9
4	5	6	10
7	8		

```
\Abacus{4}{4^3,2,1^2,0}
\Tableau{1239,456{10},78}
```

1.1

If you already know what tableaux and abacuses are, then you can probably work out the (basic) syntax of these two commands from this example and start using the package. (If you do not know what tableaux and abacuses are, then you probably don't need this package!) For more advanced usage, such as adding styling, I recommend skimming through the manual.

To use the package, add the following line to your document preamble:

```
\usepackage[options]{atableau}
```

Here, the *options* is an optional comma-separated list of **aTableau** package options — we describe the possible options below. For example, to make the tableaux and Young diagrams commands use the **French** convention, by default, you could use:

```
\usepackage[french]{atableau}
```

You can change the default options at any point in your document using the `\aTabset` command. Options can also be applied to individual commands.

The full list of commands provided by the **aTableau** package is:

Command	Diagram	Section
<code>\Abacus</code>	An abacus for a partition	Chapter 3
<code>\Diagram</code>	A Young diagram for a partition	Section 2B
<code>\Multidiagram</code>	An ℓ -tuple of Young diagrams	Section 2G
<code>\Multitableau</code>	An ℓ -tuple of tableaux	Section 2G
<code>\RibbonTableau</code>	A ribbon tableau	Section 2F
<code>\ShiftedDiagram</code>	A shifted Young diagram	Section 2D
<code>\ShiftedTableau</code>	A shifted tableau	Section 2D
<code>\SkewDiagram</code>	A skew Young diagram	Section 2C
<code>\SkewTableau</code>	A skew tableau	Section 2C
<code>\Tableau</code>	A tableau	Section 2A
<code>\Tabloid</code>	A tabloid	Section 2E

(In addition, the `\aTabset` command can be used to set global **aTableau** options.)

Each of these commands is designed to be easy to use and easy to read, and it is highly customisable. Each command accepts an optional list of key-value options that modify the diagrams. In addition, optional styling can be applied to entries of a tableau and the beads in an abacus. Each command can be used as a standalone object in a document, or they can be incorporated into a larger **tikzpicture** environment, making makes it possible to use these commands as part of more complicated diagrams.

The general syntax of an **aTableau** command is:

```
\Command (x,y) [options] {...specifications...}
```

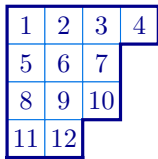
The (x,y) -Cartesian coordinates are optional, being necessary if (and only if), the diagram is part of a **tikzpicture** environment. Similarly, the key-value *options* control the style and appearance of the final diagram.

The commands are intended to be easy to use. For example, partitions can be entered either as a (weakly decreasing list of non-negative) integers, or using exponential notation, or as a sequence of beta numbers (for the `\Abacus` command), and tableaux are entered as comma-separated words. The commands support the main conventions for drawing tableaux and diagrams. By way of example, the following table shows how to draw a Young diagram of shape $\lambda = (4, 3^2, 2)$ using the four tableaux conventions supported by **aTableau**:

Convention	aTableau command	Diagram
English	<code>\Tableau[english]{1234,567,89{10},{11}{12}}</code> (the default)	
French	<code>\Tableau[french]{1234,567,89{10},{11}{12}}</code>	
Ukrainian ¹	<code>\Tableau[ukrainian]{1234,567,89{10},{11}{12}}</code>	
Australian ²	<code>\Tableau[australian]{1234,567,89{10},{11}{12}}</code>	

This table shows how the commands provided by **aTableau** work: there are basic commands for drawing diagrams, and these diagrams can be embellished using optional arguments, which use a key-value interface.

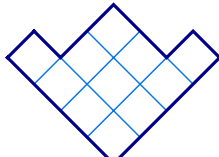
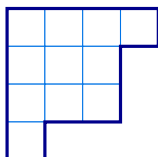
The different conventions for diagrams, and tableaux, are listed in order of (my perception of) their popularity in the literature. By default, the *English* convention is used, so we can omit *english*:



```
\Tableau{1234,567,89{10},{11}{12}}
```

1.2

for the first tableau in the table. These four conventions are used in exactly the same way with the `\Diagram` command:



```
\Diagram{4,3,3,1}
\Diagram[ukrainian]{4,3^2,1}
```

1.3

This example shows that partitions can either be typed out in full, as a comma-separated list, or using exponential notation, which is usually shorter and easier to read.

The possible options, or keys, for the **aTableau** commands are described in detail below, with examples. This section emphasises only some of these options together with the fact that every key can be set “globally” (inside the current L^AT_EX group), by using `\aTabset{...options...}`. As discussed already, the default options for the document can also be set via `\usepackage[options]{atableau}`.

Many of the **aTableau** options are specific to the particular diagrams being drawn. The following options are common to all of the commands.

baseline

Sets the baseline for vertically aligning diagrams. The possible options are `align=top`, `align=center`, and `align=bottom`, which aligns the baseline of the diagram with the top, centre or bottom of the surrounding objects, respectively. By default, **aTableau** are centered.

name

Set the prefix for the **TikZ** *named anchors* for the boxes and beads in **aTableau** diagrams.

¹Some authors call this the Russian convention.

²I have not seen a name for this convention in the literature. Calling this the *Australian convention* seems apt.

scale, xscale, yscale :

Use **scale** to rescale all **aTableau** diagrams. Use **xscale** to rescale in the x -direction and **yscale** to rescale in the y -direction.

star style :

Adding a $*$ -prefix to elements of an **aTableau** diagram is a shorthand for changing the style of that element. Use **star style** to change the default $*$ -styling.

tikzpicture :

All **aTableau** commands are drawn inside a **tikzpicture** environment. Use **tikzpicture=keys** to set the optional argument of this environment:

```
\begin{tikzpicture}[ keys ]...\end{tikzpicture} .
```

tikz before, tikz after :

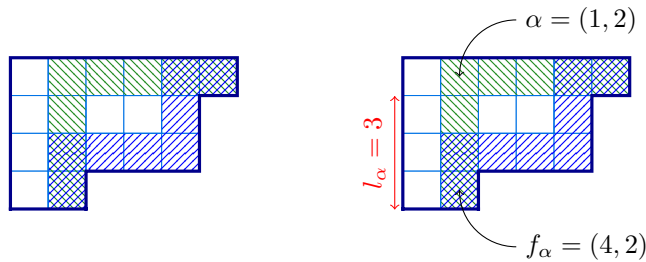
Use the **tikz before** and **tikz after** keys to inject **TikZ** code into the **tikzpicture** environment, before and after the diagram drawn by **aTableau**.

With boolean options, which are set to true or false, the value can typically be omitted. For example, **border** is the same as **border=true** (and **no border** is the same as **border=false**). In all of the options, American spelling variations, such as *center* (instead of *centre*), *color* (instead of *colour*) et cetera, are tolerated. All of the colours in **aTableau** diagrams can be changed using the key-value options. The names of the underlying base colours used by **aTableau** are given in [Section 3C](#).

The next example shows how, how to use the **\Diagram** command inside a **tikzpicture** environment by supplying Cartesian coordinates.

```
% \usetikzlibrary{patterns}
\tikzset{
  B/.style={pattern=north east lines, pattern color=blue},
  G/.style={pattern=north west lines, pattern color=ForestGreen},
}
\Diagram[ribbons={{(G)16ccccrrr, (B)16crrcccr}}{6,5^2,2} % unlabelled diagram

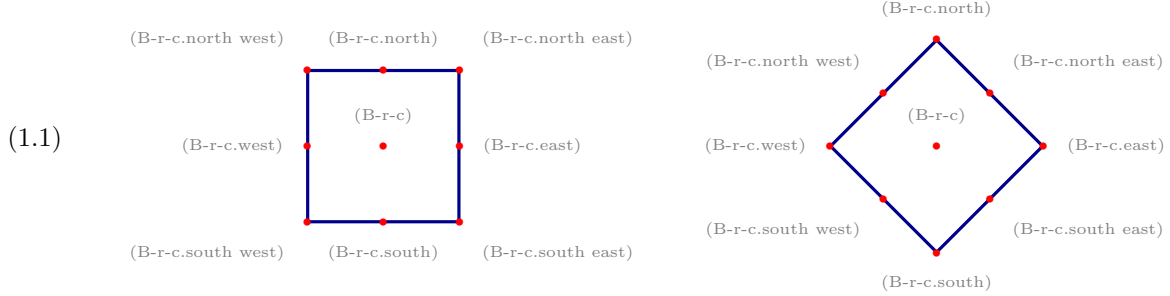
\begin{tikzpicture} % with labels using a tikzpicture environment
  \Diagram(0,0)[ribbons={{(G)16ccccrrr, (B)16crrcccr}}{6,5^2,2}
  % add the labels
  \draw[->](1.5, 0.5)node[right]{ $\alpha=(1,2)$ } to [out=180,in=90] (B-1-2.base);
  \draw[->](1.5,-2.5)node[right]{ $f_\alpha=(4,2)$ } to [out=180,in=270] (B-4-2.base);
  \draw[red,<->]([xshift=-1mm]B-1-1.south west)
    --node[rotate=90,above]{ $l_\alpha=3$ }([xshift=-1mm]B-4-1.south west);
\end{tikzpicture}
```



The diagram is drawn twice, both outside and inside a **tikzpicture** environment. The first diagram, which does not have labels, emphasizes that most of the right-hand diagram is drawn by the **\Diagram** command, even though most of the commands in the example are **TikZ** commands. In particular, the shading of the boxes in the $(1,2)$ -hook and $(1,2)$ -rim hook is done by the **ribbons** key. (In fact, this diagram could be drawn without using a **tikzpicture** environment by adding the **\draw**-commands to the **tikz after** key.)

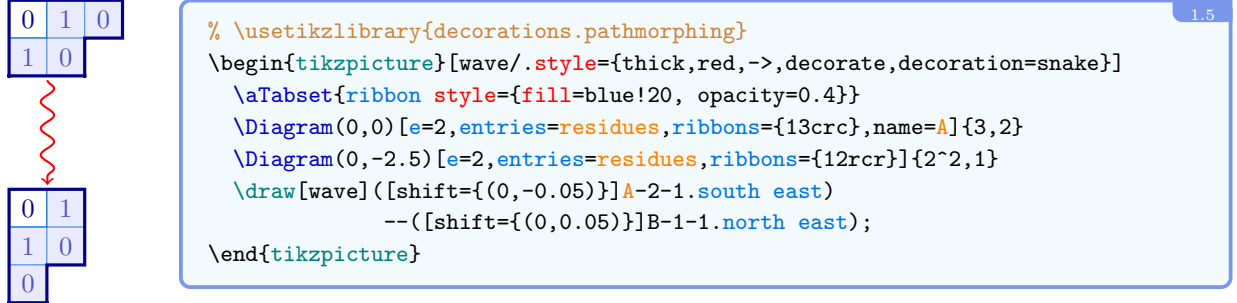
The top of the last example defines two **TikZ**-styles, **B** and **G**, which add the blue and green coloured hatchings of the two hooks, respectively. Inside the **tikzpicture** environment, the Young diagram is placed at position $(0,0)$ by the **\Diagram** command. The three **\draw** commands are used to add the three labelled arrows to the picture. Please consult the very readable **TikZ** manual for more details on the **TikZ**-commands in this example.

From the viewpoint of **aTableau**, the most important point of the last example is that the first `\Diagram` command is a standalone diagram, whereas the second one is part of a `tikzpicture` environment because it is equipped with the Cartesian coordinate (0,0). The second important point is that this example uses four *named coordinates*: (B-1-1), (B-1-2), (B-4-1), and (B-4-2). When **aTableau** draws a tableau, or a diagram, it creates a **TikZ** named coordinate (B-r-c) for every box in the diagram, where (B-r-c) is the box in row *r* and column *c*. These named coordinates, or anchors, are *only* defined for the boxes in the tableau. Giving finer control, the following standard **TikZ** anchors are available:

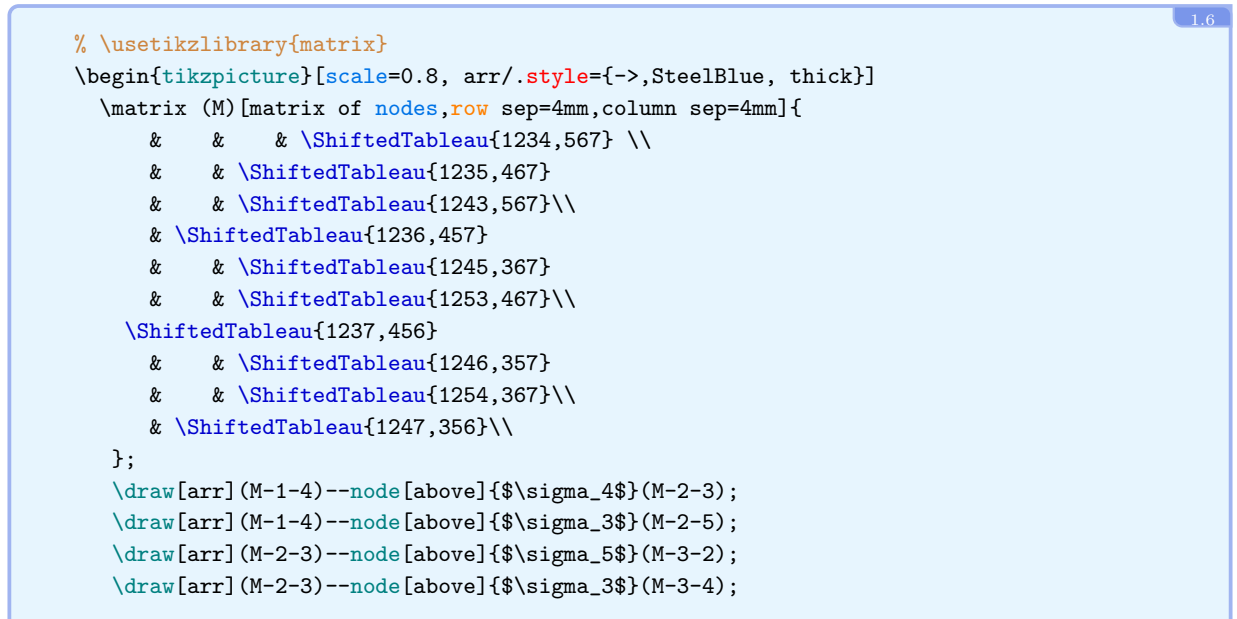


You can change the B in the name of these anchors to anything you like using the `name` key. Similar anchors are created for abacus beads. These *anchors* are a standard part of **TikZ**. If you change the shape of the node, then the available anchors may change; for more information see the **TikZ** manual.

In the example above, the Cartesian coordinate (0,0) is necessary because the `\Diagram` command is used inside a `tikzpicture` environment, but the choice of coordinate is not important in the sense that changing the coordinate will not change the picture. Cartesian coordinates become more meaningful when the `tikzpicture` environment has more components. In the next example, notice the use of `name=A` to make the named coordinates for the first diagram take the form (A-r-c), instead of the default node names (B-r-c), which are used for the second diagram.



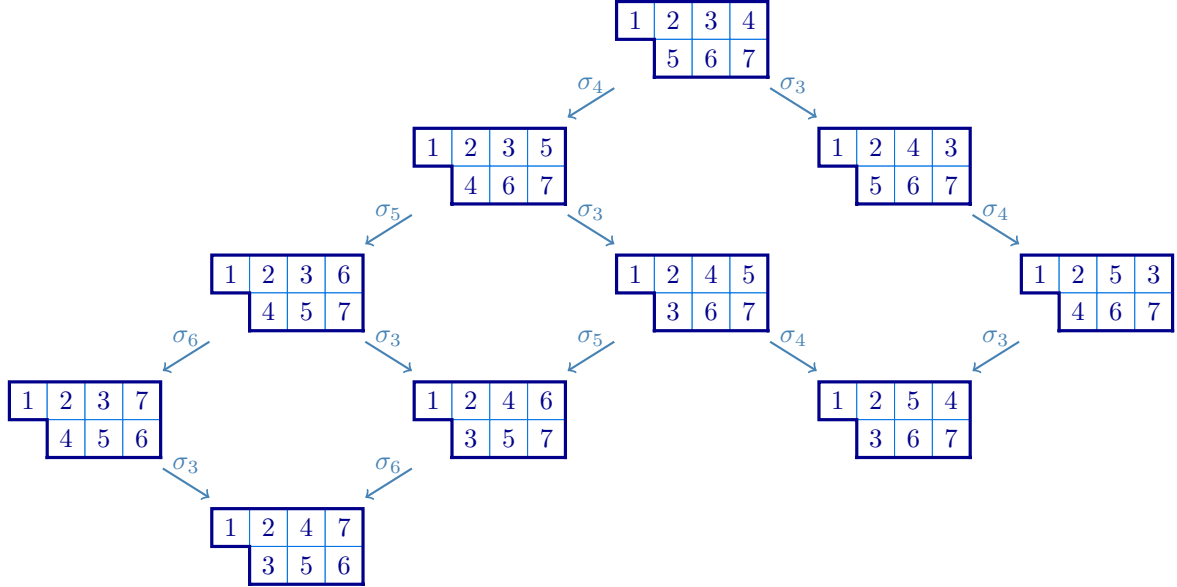
The final example in the introduction uses `\ShiftedTableau` inside a *matrix of nodes*:



```

\draw[arr] (M-2-5)--node[above]{\sigma_4}(M-3-6);
\draw[arr] (M-3-2)--node[above]{\sigma_6}(M-4-1);
\draw[arr] (M-3-2)--node[above]{\sigma_3}(M-4-3);
\draw[arr] (M-3-4)--node[above]{\sigma_5}(M-4-3);
\draw[arr] (M-3-4)--node[above]{\sigma_4}(M-4-5);
\draw[arr] (M-3-6)--node[above]{\sigma_3}(M-4-5);
\draw[arr] (M-4-1)--node[above]{\sigma_3}(M-5-2);
\draw[arr] (M-4-3)--node[above]{\sigma_6}(M-5-2);
\end{tikzpicture}

```



Notice that even though the `\SkewTableau` commands are being used inside a `tikzpicture` environment they do not need coordinates because the skew tableaux are not explicit components in the environment, being all (implicitly) inside `TikZ`-nodes.

The following sections describe the commands defined by the `aTableau` package and how to use them. We start with the `\Tableau` command because all of the diagram and tableau commands, except for `\RibbonTableau`, are derived from `\Tableau`.

2. TABLEAUX AND YOUNG DIAGRAMS

Partitions, which are weakly decreasing sequences of non-negative integers, are fundamental objects in algebraic combinatorics and representation theory. Rather than working with sequences of integers, it is often useful to visualise partitions as Young *diagrams*, which are arrays of *boxes*, or *nodes*, in the plane³. A *tableau* is a (Young) diagram where the boxes are labelled using some alphabet. Equivalently, a diagram is a tableau with empty labels.

For example, $\lambda = (4, 3, 3, 2, 0, 0, \dots)$ is a partition of 12, since the entries sum to 12. It is customary to omit the zeros when writing partitions and to use exponents for repeated parts, so we can write this partition in *exponential notation* as $\lambda = (4, 3^2, 2)$. Many authors identify a partition $\lambda = (\lambda_1, \lambda_2, \dots)$ with its *diagram*, which is the set $[\lambda] = \{(r, c) \mid 1 \leq c \leq \lambda_r \text{ for } r \geq 1\}$. When using the *English convention*, the diagram of λ is visualised as a left justified array of boxes in the plane, where the first row has 4 boxes, the next two lower rows have 3 boxes and the last row has 2 boxes. The *shape* of a diagram is the partition that gives the number of boxes in each row. Diagrams and tableaux drawn using the *French convention* are given by reflecting in a horizontal line above the diagrams, whereas the *Ukrainian* and *Australian* conventions are given by appropriately rotating these diagrams. All of these conventions are used in the literature, with some papers using more than one convention.

³In this manual, we normally refer to the entries of tableaux and diagrams as *boxes*, rather than *nodes*, to avoid any ambiguity with `TikZ`-nodes.

2A. **Tableaux.** This section describes the `\Tableau` command, which draws tableaux or labelled diagrams. The syntax of this command is:

`\Tableau (x,y) [options] {tableau entries}`

where:

(x,y) : The Cartesian coordinates (x,y) are needed if, and only if, the tableau is a component of a `tikzpicture` environment.

options : Optional arguments. Described, with examples, below.

tableau entries : The `tableau entries` are given as a comma-separated list, with commas separating rows and each *token*, or braced-group of tokens, being in a separate column. As we explain below, each tableau entry can be prefixed by (optional) `TikZ`-styling specifications.

We show by example how to use the `\Tableau` command, starting with basic usage and finishing with style. Inside `\Tableau`, the rows are separated by commas, with the columns given by the “letters” in the word for each row:

1	2	3	4
5	6		
7			
8			

α	γ	δ	q
x	y	η	
λ	μ		
π	$\sum$		

```
\Tableau{ 1234, 56, 7, 8 }
\Tableau{ \alpha\gamma\delta q,
           xy\eta,
           \lambda\mu,
           \pi\sum }
```

2A.1

The tableau specifications can be put on a single line, without any spaces. Alternatively, as shown here, spaces and line breaks can be added to improve readability. The one restriction is that you cannot have blank lines inside a `\Tableau` command. As this example suggests, by default, the entries in a tableau are typeset in mathematics-mode, but this can be changed using the `text boxes` key, which is described below.

As the examples above show, the tableau entries can be individual characters, or they can be $\text{\TeX}$ commands. Tableaux frequently contain entries that are not single characters, or given by commands. Such entries can be added to a tableau by enclosing them in braces:

1	3	10	11	2x
2	x^2			

```
\Tableau{ 13{10}{11}{2x}, 2{x^2} }
```

2A.2

There is one additional caveat for braced-entries: any *singleton* braced entry `{...}` in a row needs to be *double-braced*. Double-bracing is only necessary when you have a row that has only one column, and you have a multi-letter column entry. The need for double-bracing is a consequence of how $\text{\LaTeX}$ 3 detects braced commas in comma-separated lists.

The following example shows the difference between single and double-bracing for tableau entries.

1	3	4	5
8	10		
1	1		
1	2		

1	3	4	5
8	10		
11			
12			

```
\Tableau{ 1345, 8{10}, {11}, {12} }
\Tableau{ 1345, 8{10}, {{11}}, {{12}} }
```

2A.3

The 10's in these tableaux do not need to be doubled-braced because they appear in a row of length 2, which is greater than 1!. In contrast, because they are not doubled-braced, the 11 and 12 in the first tableau are both treated as being two characters in different columns in the first tableau. In the second tableau, the 11 and 12 each appear in a separate column because they are double-braced.

2A.1. *Adding style.* The `aTableau` package provides several different ways to add `TikZ` style specifications to the entries in tableaux, and to boxes in diagrams. This is made possible by adding style prefixes to the tableau entries. To take advantage of these styles you need to have at least rudimentary knowledge of `TikZ`-styling. The examples in this manual give a good indication of what is possible. For anything more exotic, please consult the `TikZ` manual.

The simplest way to change the style of an entry in a tableau is to add a *-prefix to the entry:

1	2	3	4	5
6	7	8	9	
10	11	12		
13				

```
\Tableau{ 1*2*3*4*5, 6*789, {10}*{11}{12}, {{13}} }
```

2A.4

By default, the *-entries are given the [TikZ](#)-styling `fill=` . You can change the default *-style using the [star style](#) key:

1	2	3	4	5
6	7	8	9	
10	11	12		
13				

```
\Tableau[star style={fill=red!20,draw=red,thick}]
{ 1*2*3*4*5, 6*789, {10}*{11}{12}, {{13}} }
```

2A.5

Note that the *-styling overrides the default styling of the surrounding boxes. This is because the boxes with the default styling are placed first, in the order that they are entered, followed by boxes with custom styles are placed.

The *-syntax is a quick shorthand for adding emphasis to some boxes in a tableau, but each of these entries will be given the same style. It is possible to give every box in the tableau different styles by putting [TikZ](#)-styling specifications inside square brackets, `[...]`, before the corresponding letter in the tableau specification. You can mix the *-syntax and the `[...]`-style syntax for different entries, although only one of these style settings can be applied to any given box.

13				
10	11	12		
6	7	8	9	
1		3		

```
\Tableau[french]{
  1 [blue!20]2 [circle]3 [red]4 [cyan]5,
  6 *7 8 9,
  {10} *{11} {12},
  {{13}} }
```

2A.6

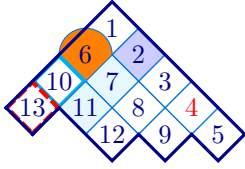
It is not necessary to add spaces between the entries in the way that this example does. This is done only to make it clearer how the style specifications are applied to the different entries in the tableau.

Omitting some technicalities, each box in a tableau is constructed as a [TikZ](#) node, so any [TikZ](#)-styling specifications for a node can be used. This example shows that some care is needed when changing the style because simply giving a colour changes both the colour used to fill the box and the text colour, which is why some of tableau entries in the last example appear to be blank. If you are already familiar with [TikZ](#) then you probably already know what to do. If not, then here is a short list of useful style specifications for [TikZ](#) nodes:

Style	Meaning
<code>draw = <colour></code>	Sets the boundary colour of the node
<code>fill = <colour></code>	Sets the fill, or background, colour of the node
<code>font = </code>	Sets the font
<code>text = <colour></code>	Sets the text colour in the node

In the style settings, any valid $\text{\LaTeX}$ colour name can be used; see, for example, the [xcolor](#) manual. In addition, the style specifications `thin`, `thick`, `very thick`, `ultra thick`, `dashed`, `dotted`, ... change the border thickness, and its properties. [Subsection 2A.3](#) below describes how to use these options to customise the tableau style.

If you want to apply “complicated” style settings, such as `draw=cyan,ultra thick`, which consists of a comma-separated list of [TikZ](#)-style settings, then the entire style must be put inside matching square brackets and curly braces `[{...}]`. This is necessary to ensure that $\text{\LaTeX}$ does not get confused by the commas in the style setting (since commas are also used to separate the rows of the tableau).



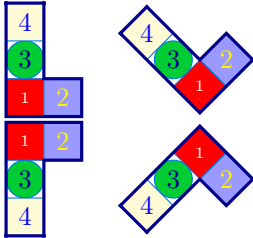
```
\Tableau[australian]{
  1 [fill=blue!20]2 3 [text=red]4 5,
  [{circle,fill=orange}]6 *7 8 9,
  [{draw=cyan,ultra thick}]{10} *{11} {12},
  [{dashed,draw=red,ultra thick}]{13} }
```

2A.7

The last example shows that it is not necessary to double-brace singleton entries when they have a style specification.

This example shows that modifying the borders of a box is problematic because boxes placed later are likely to overwrite the border (see Subsection 2A.5). As a result, modifying the borders of the boxes is not a good way to add emphasis to a tableau, so use this sparingly. In addition, the orange circle in this example looks too big, but it is what we asked for because the diameter of the circle is the horizontal width of the box. The next example shows that we get a better result for *Australian* and *Ukrainian* tableaux if we add a *minimum size* specification.

For complex or frequently used styles, we recommend using the `\tikzset` command to define the style outside of the `\Tableau` command. This makes the style both easier to read and easier to change.

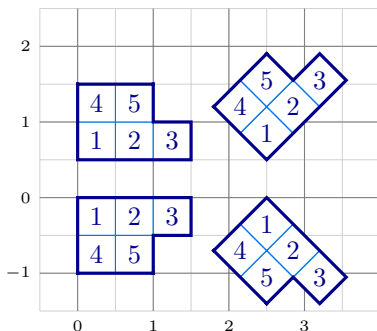


```
\tikzset{
  B/.style={fill=blue!40, text=yellow},
  G/.style={fill=green!80!blue,circle,minimum size=5mm},
  R/.style={fill=red,text=white, font=\tiny},
  Y/.style={fill=yellow!20, text=blue}
}
\Tableau[french]{[R]1[B]2,[G]3,[Y]4}
\Tableau[ukrainian]{[R]1[B]2,[G]3,[Y]4}
\Tableau[english]{[R]1[B]2,[G]3,[Y]4}
\Tableau[australian]{[R]1[B]2,[G]3,[Y]4}
```

2A.8

In this example, the `TikZ`-styles B, G R, and Y are applied to the tableaux entries labelled 1, 2, 3, and 4, respectively. Note the use of *minimum size* to control the size of the circle. The colours used here, and throughout this manual, are for demonstration purposes only. I recommend against such garish choices in any self-respecting document!

2A.2. Tableau coordinates. The (x, y) -coordinates are required if, and only if, the `\Tableau` command is used inside a `tikzpicture` environment, in which case (x, y) gives the coordinates of the “outside corner” of the box in row 1 and column 1 of the tableau. The following example shows how tableaux are placed using (x, y) -coordinates inside a `tikzpicture` environment. The example also shows that, by default, the tableau boxes are square and half a unit wide.



```
\begin{tikzpicture}[add grid]
  \Tableau(0,0) [english] {123,45}
  \Tableau(0,0.5) [french] {123,45}
  \Tableau(2.5,0.5) [ukrainian] {123,45}
  \Tableau(2.5,0) [australian]{123,45}
\end{tikzpicture}
```

2A.9

2A.3. Tableau options. The optional `*-styles` and `[...]-styles` are the main mechanism for styling the individual boxes in a tableau. Using a key-value syntax, you can change aspects of the tableaux produced by `\Tableau` commands. The rest of this section describes these keys, their default values, and gives examples of how to use them.

The options below can be applied to the `\Tableau` command by giving them as a comma-separated list inside square brackets:

`\Tableau [options] {tableau entries}` or `\Tableau (x,y) [options] {tableau entries}`.

The options can appear in any order, with later options having precedence over earlier ones.

When used inside a `\Tableau` command, the options only affect that particular tableau. Most of these options, or settings, can also be used in the `\aTabset` command, or as optional arguments in `\usepackage[...]{atableau}`, to change the default settings of all tableaux in the same L^AT_EX group. For example, if you put the command `\aTabset[ukrainian]` in the preamble of your document then, by default, all subsequent tableaux and Young diagrams will be drawn using the Ukrainian convention.

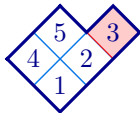
`english` (default: `english`)

`french`

`ukrainian`

`australian`

These four (mutually exclusive) options change the convention, or orientation, that is used when drawing the tableaux. Rather than defining these conventions precisely, we use the tired and true method in combinatorics of defining by example.



```
\tikzset{R/.style={fill=red!20,draw=red}}
\Tableau[ukrainian]{12[R]3,45}
```

2A.10



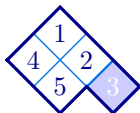
```
\tikzset{R/.style={fill=red!20, circle, draw=red, ultra thick}}
\Tableau[french]{12[R]3,45}
```

2A.11



```
\tikzset{B/.style={fill=blue!20, dashed, thick}}
\Tableau[english]{12[B]3,45}
```

2A.12



```
\tikzset{B/.style={fill=blue!20, thick, text=white}}
\Tableau[australian]{12[B]3,45}
```

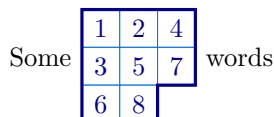
2A.13

By default, the `english` convention is used to draw tableaux and Young diagrams. The default convention can be changed using `\aTabset`. Capitalised names, `Australian`, `English`, `French` and `Ukrainian`, are also recognised. If, for example, your document is only going to draw `french` tableaux and diagrams, then you can specify this when you load the package using `\usepackage[french]{atableau}`, or by adding `\aTabset{french}` to your document preamble.

`align` (default: `centre`)

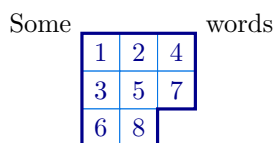
[accepts: `centre`/`bottom`/`top`]

By default, the baseline of a tableau is its centre. This can be changed using the `align`.



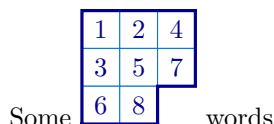
```
Some \Tableau[align=centre]{124,357,68} words % the
default
```

2A.14



```
Some \Tableau[align=top]{124,357,68} words
```

2A.15



```
Some \Tableau[align=bottom]{124,357,68} words
```

2A.16

Similarly, use `align` to control how tableaux are aligned in displayed equations.

2A.17

```
\aTabset{align=bottom}
\Tableau{124,357,68}\cquad
\Tableau{124,35}
```

1	2	4
3	5	7
6	8	

1	2	4
3	5	

2A.18

```
% by default: align=centre
\Tableau{124,357,68}\cquad
\Tableau{124,35}
```

1	2	4
3	5	7
6	8	

1	2	4
3	5	

2A.19

```
\aTabset{align=top}
\Tableau{124,357,68}\cquad
\Tableau{124,35}
```

1	2	4
3	5	7
6	8	

1	2	4
3	5	

`border` (default: `true`)
`no border` (default: `false`)

[accepts: `true/false`]
[accepts: `false/true`]

The `border` and `no border` options enable and disable the drawing of the (thick) border wall around a tableau. As with other boolean keys, `border` is equivalent to `border=true` and `no border=false`. The `no border` key is inverse to `border`, so `no border` is equivalent to `no border=true`, and to `border=false`.

	4	
7	5	
6	5	3
4	2	
	1	

6	7	4
4	5	5
1	2	3

2A.20

```
\Tableau[ukrainian, border=false]{123,455,674}
\Tableau[french, no border]{123,455,674}
```

`border colour` (default:)
`border style` (default: `thick,line cap=rect`)

[accepts: any colour]
[accepts: `TikZ`-styling]

The `border colour` option sets the colour of the border wall, whereas `border style` sets the `TikZ`-styling of the border wall. (So, `border style` can override `border colour`.)

	4	
7	5	
6	5	3
4	2	
	1	

4	5	5
1	2	3

2A.21

```
\Tableau[ukrainian, border colour=red]{123,455,674}
\Tableau[french, border style={dashed,orange}]{123,455}
```

The `border style` key *appends* any `TikZ`-styles to the current styling of the wall.

`box height` (default: `0.5`)
`box width` (default: `0.5`)

[accepts: a decimal giving the height in cm]
[accepts: a decimal giving the width in cm]

aTableau

Sets the width and height of the boxes in the tableau. The default height and width of the boxes is 0.5 cm when using the [english](#) and [french](#) conventions and 0.7012 cm, which is approximately $1/\sqrt{2}$ cm, when using the [ukrainian](#) and [australian](#) conventions. (In all cases, the side length of the squares is 0.5 cm.)

1	2	3
4	5	5
6	7	4

	5	
5		3
4	2	
	1	

```
\aTabset{align=top}
\Tableau[box height=0.5]{123,455,674}
\Tableau[box width=0.2,ukrainian]{123,455}
```

2A.22

See also the [scale](#), [xscale](#) and [yscale](#) options.

box fill (default: [none](#))

[accepts: any colour]

The **box fill** option sets the background colour of a box in a tableau, just like the fill key for a [TikZ](#)-node. If you use a dark fill colour, then you will almost certainly want to change the text colour using the **text** option – and you may also want to change the colours of the border walls and inner walls using **border colour** and **inner colour**, respectively.

1	2	3
4	5	5
6	7	4

1	2	3
4	5	5

```
\Tableau[box fill=blue,box text=yellow]{123,455,674}
\Tableau[box fill=red!10]{123,455}
```

2A.23

box font

[accepts:]

The **box font** key sets the font used to type the entries of a tableau. By default, tableau entries are typeset as mathematics, so this is mainly useful for changing the font size (because, for example `\bfseries $1$` does not make the 1 bold). It is only when you are using **text boxes** that font commands like `\itshape` and `\bfseries` will have any effect.

1	2	3
4	5	5
6	7	4

1	2	3
4	5	5

```
\Tableau[box font=\tiny]{123,455,674}
\Tableau[text boxes, box font=\bfseries]{123,455}
```

2A.24

box text (default: [black](#))

[accepts: any colour]

As for a [TikZ](#)-nodes, use **box text** to set the default text colour for the entries of the tableau boxes.

1	2	3
4	5	

4	5	
1	2	3

```
\Tableau[box text=red]{123,45}
\Tableau[box text=blue, french]{123,45}
```

2A.25

conjugate (default: [false](#))

[accepts: false/true]

Draws the *conjugate* tableau, where the rows and columns are swapped.

1	2	3
4	5	
6	7	

1	4	6
2	5	7
3		

```
\Tableau{123,45,67}
\Tableau[conjugate]{123,45,67}
```

2A.26

halign (default: [centre](#))

[accepts: centre,left,right]

valign (default: [centre](#))

[accepts: bottom,centre,top]

By default, the entries in tableaux are centered, both horizontally and vertically, and it is your responsibility to ensure that the entries fit inside the tableau boxes (you can change the box dimensions using the **box height**, **box width** and **scale** keys). The **halign** and **valign** keys provide a crude way to change the horizontal and vertical alignment of the box entries.

1	11	11
1	1	

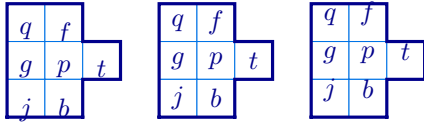
1	11	11
11	1	

1	1	1
1	1	

```
\Tableau[halign=left]{1{11}{11},{11}1}
\Tableau[halign=centre]{1{11}{11},{11}1}
\Tableau[halign=right]{1{11}{11},{11}1}
```

2A.27

aTableau



```
\Tableau[valign=bottom]{qf,gpt,jb}
\Tableau[valign=centre]{qf,gpt,jb}
\Tableau[valign=top]{qf,gpt,jb}
```

2A.28

These keys are provided for completeness only. We recommend not using them!

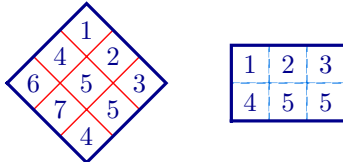
inner colour (default:)

[accepts: any colour]

inner style (default: `thin,line cap=rect`)

[accepts: **TikZ**-styling]

The **inner colour** option sets the colour of the inner wall, whereas **inner style** sets the **TikZ**-styling of the inner wall. (In particular, **inner style** can be used to override **inner colour**.)



```
\Tableau[australian, inner colour=red]{123,455,674}
\Tableau[inner style=dashed]{123,455}
```

2A.29

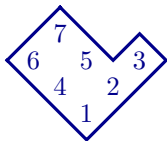
If you closely at the second tableau in [Example 2A.32](#), you will notice that the dashes are not very clean. This is because, for example, the dashes on bottom of one box do not properly match up with the dashes on the top of the adjacent box below it. Rather than using **inner style=dashes** it is better to use something like **inner style={dash pattern}**.



```
\Tableau[inner style={dash pattern=on 1pt off 2pt}]
{123,455}
```

2A.30

Use **inner colour=none**, to remove the inner tableau walls.



```
\Tableau[ukrainian, inner colour=none]{123,45,67}
```

2A.31

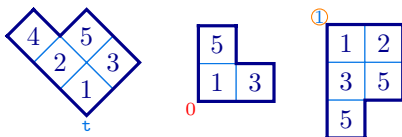
label

[accepts: any character/text]

label style (default: `font=\scriptsize,text=` `)`

[accepts: **TikZ** commands]

The **label** key adds a label to a tableau next to the (1,1)-box. The style of the label can be changed using the **label style** key.



```
\Tableau[ukrainian, label={\mathtt{t}}]{13,25,4}
\Tableau[french,label=0,label style={red}]{13,5}
\Tableau[label style={draw=orange,circle,inner sep=0pt},
label=1]{12,35,5}
```

2A.32

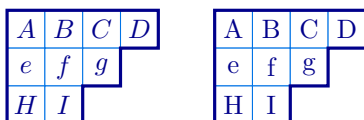
math boxes (default: `true`)

[accepts: true/false]

text boxes (default: `false`)

[accepts: true/false]

Use the **math boxes** and **text boxes** to have the tableau entries typeset as either mathematics or text, respectively. By default, the entries of a tableau are typeset as mathematics.



```
\Tableau[math boxes]{ABCD, efg, HI}
\Tableau[text boxes]{ABCD, efg, HI}
```

% the default

2A.33

name (default: `B`)

[accepts: string]

By default, the boxes can be referenced using the node names (B-1-1), (B-1-2), (B-1-3), ..., with (B-*r*-*c*) referring to the box in row *r* and column *c*. This feature is most useful when you have several tableaux inside a **tikzpicture** environment because if you only have one tableau, then it is unlikely that you need

to change the default prefix for the node names. See (1.1) for a description of the anchor names, as well as Example 1.5 and Example 2A.46 for examples using node names and anchors.

ribbons [accepts: list of ribbon specifications]
snobs [accepts: list of ribbon specifications]
ribbon style [accepts: **TikZ**-styling commands]

Use the **ribbons** key to add a comma-separated list of ribbons to a tableau. Use **snobs** to add ribbons to the tableau that are placed *after* the border of the diagram is drawn. (Writing *ribbons* backwards gives *snobbir*.) The **ribbon style** key controls the style of the ribbons.

When adding ribbons, or snobs, you first specify the row and column indices of the *head* of the ribbon, which is the unique node of maximal content in the ribbon, and then give a sequence of *r*'s and *c*'s depending on whether the ribbon moves along a row or column. As always, you have the option of adding style — and you can also specify the contents of each box in the ribbon. For more details about the ribbon specifications, with examples, see Section 2F, which describes the `\RibbonTableau` command.

Use the **ribbon style** option to change the default style of the ribbons. For example, noting that a box is a ribbon of length 1, the following code uses ribbons of length one (that is, nodes), to highlight the addable nodes of this tableau.

1	2	3	4	A
5	6	7	B	
8	9	10		
11	C			
D				

```
\Tableau[star style={fill=green!80!blue,text=white},
ribbon box={fill=yellow,text=red},
ribbons={15_A,24_B,42_C,51_D},
]{123*4,567,89*{10},*{11}}
```

2A.34

As this example indicates, ribbons are assumed to be contained inside the diagram of the tableau, so they do not change the border of a tableau.

The default style for a **snobs** ribbon is given by **ribbon style**.

1	2	3	4	A
5	6	7	B	
8	9	10		
11	C			
D				

```
\Tableau[star style={fill=green!80!blue,text=white},
ribbon box={fill=yellow, text=red},
snobs={15_A,24_B,42_C,51_D},
]{123*4,567,89*{10},*{11}}
```

2A.35

The difference between these two pictures is that in the second picture the nodes in the **snobs** are drawn over the tableau border, causing the border to disappear for these nodes. For this reason, in most cases you should use **ribbons**, and avoid **snobs**.

scale (default: 1) [accepts: a decimal number]
xscale (default: 1) [accepts: a decimal number]
yscale (default: 1) [accepts: a decimal number]

By design, boxes in tableaux do not automatically resize to fit their contents. For example, consider:

1	3	10	11	+	x
2	+	x			

```
\Tableau{ 13{10}{11}{1+x}, 2{1+x} }
```

2A.36

This is unlikely to be the desired output! The options **scale**, **xscale** and **yscale** can be used to rescale the boxes in tableaux and diagrams. The names are slightly misleading because **xscale** rescales in the direction of increasing column index and **yscale** rescales in the direction of increasing row index.

1	3	10	11	+	x
2	+	x			

```
\Tableau[scale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```

2A.37

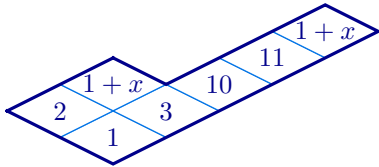
aTableau

1	3	10	11	$1+x$
2	$1+x$			

```
\Tableau[xscale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```

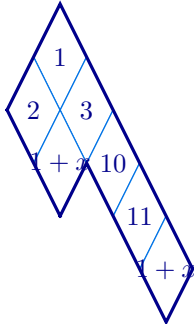
2A.38

When using the [australian](#) and [ukrainian](#) conventions, [xscale](#) and [yscale](#) scale in both the x direction and the y-direction, reflecting how these tableaux grow with increasing column and row index, respectively.



```
\Tableau[ukrainian, xscale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```

2A.39



```
\Tableau[australian, yscale=2]{ 13{10}{11}{1+x}, 2{1+x} }
```

2A.40

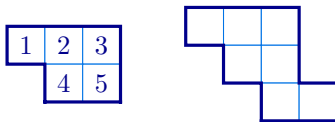


The options, [scale](#), [xscale](#), and [yscale](#), affect all [aTableau](#) diagrams, including tableaux, Young diagrams and abacuses. If you want to use [\aTabset](#) to change the default sizes of the diagrams and tableaux in your document, then it is probably better to use the [box height](#) and [box width](#) options.

[shifted](#) (default: [false](#))

[accepts: true/false]

Set to [true](#) for a shifted tableau or shifted diagram. Shifted tableaux, and shifted diagrams, are discussed in more detail in [Section 2D](#), which describes the [\ShiftedDiagram](#) and [\ShiftedTableau](#) commands.



```
\Tableau[shifted]{123,45}
\Diagram[shifted]{3,2^2}
```

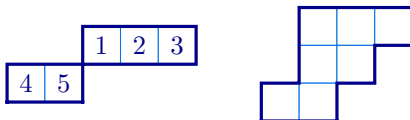
2A.41

See [Section 2D](#) for the options specific to shifted tableaux and shifted diagrams.

[skew](#) (default: [\(0\)](#))

[accepts: a partition]

Specify the skew shape for a skew tableau or skew diagram. Skew tableaux, and skew diagrams, are discussed in more detail in [Section 2C](#), which describes the [\SkewDiagram](#) and [\SkewTableau](#) commands.



```
\Tableau[skew={2}]{123,45}
\Diagram[skew={2^2,1}]{3,2^2}
```

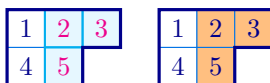
2A.42

See [Section 2C](#) for the options specific to skew tableaux and skew diagrams.

[star style](#) (default: [fill=](#) )

[accepts: [TikZ](#)-styling]

Use the [star style](#) to add to the style of the starred entries of a tableau. (In [TikZ](#) parlance, [star style](#) *appends* these styles to the default [star style](#).)



```
\Tableau[star style={text=magenta,
draw=cyan, thick}]{1*2*3,4*5}
\Tableau[star style={fill=orange!50}]{1*2*3,4*5}
```

2A.43

Note that the tableau border is drawn *after* the [star style](#) is applied. You can use [snobs](#) to draw on top of the border.

`tabloid` (default: `false`)[accepts: `true/false`]

Use the `tabloid` key to draw a tabloid. This key is discussed in more detail in [Section 2E](#), which describes the `\Tabloid` command.

1	2	3
4	5	


```
\Tableau[tabloid]{123,45}
\Diagram[tabloid]{3,2^2}
```

2A.44

See [Section 2E](#) for the options specific to tabloids.

`tikz before`[accepts: `TikZ` commands]`tikz after`[accepts: `TikZ` commands]

These keys inject `TikZ` code into the underlying `tikzpicture` environment, where the tableau is constructed, before and after the `\Tableau` command, respectively.

	1	3	4	
	5	6	7	
	8	9		

```
\Tableau[
tikz before={\draw[help lines,step=0.5](-0.5,-2)grid(2,0.5);},
tikz after={\draw[ultra thick,red]
(B-1-3.north east)--(B-1-3.south east)
--(B-2-1.north east)--(B-3-1.south east)
--(B-3-1.south west);
}]{134,567,89}
```

2A.45

Both before and after hooks are provided because the order in which you place `TikZ` commands can affect the drawing, primarily because later commands are drawn over the top of earlier ones. Note that the *before* code cannot use the named nodes `B-r-c` because this code is executed before these nodes are constructed, which happens when the tableau is drawn. For more examples, see [Subsection 2A.5](#).

`tikzpicture`[accepts: `TikZ`-keys]

This command adds an optional argument to the underlying `tikzpicture` environment, which contains the tableau. The `tikzpicture` option is ignored when the `\Tableau` command is equipped with (x,y) -coordinates.

13	11	12	9	5	
	10	7	8	4	
	6	1	2	3	

```
\Tableau[ukrainian, name=A,
tikzpicture={remember picture}]
{ 1[fill=blue!20]23[text=red]45,
6*78[{\draw=cyan,ultra thick}]9,
[{\circle,fill=orange}]{10}*{11}{12},
[{\dashed,draw=red,ultra thick}]{13}
}
```

2A.46

Since `name=A`, the nodes in this tableau are referenced as $(A-1-1)$, $(A-1-2)$, $(A-1-3)$, As the example uses *remember picture*, other `tikzpicture` environments can reference the node names in last tableau, which allows us to do things like this:

This circle is too big!
Use minimum size=5mm

```
\tikz[remember picture, overlay]
\draw[very thick, ->,red]
(0,0) node[right,align=left]
{This circle is too big!\Use minimum size=5mm}
to [out=135, in=225](A-3-1);
```

2A.47

As a second example, we can use `tikzpicture={rotate=30}` to rotate a tableau by 90 degrees:

3	
2	5
1	4

3	
2	5
1	4

```
\Tableau[tikzpicture={rotate=90}]{123,45}
\begin{tikzpicture}[rotate=90]
\Tableau(0,0){123,45}
\end{tikzpicture}
```

2A.48

The two tableaux constructions in this example are equivalent. An oddity with both of these examples is that the numbers in the tableaux have not been rotated, which is because the `TikZ rotate` key does not affect component objects, such as nodes, inside a `TikZ` picture. You can rotate the entire picture using the `transform canvas` key:

3	
2	5
1	4

```
\Tableau[tikzpicture={transform canvas={rotate=90}}]{123,45}
```

2A.49

2A.4. *Compositions*. As we have defined them, tableaux are always of *partition* shape, in the sense that the lengths of the rows is a weakly decreasing sequence. In fact, the `\Tableau` command also accepts tableaux of *composition* shape, where the lengths of the rows are not necessarily in decreasing order:

4	5		
2	7	5	1
3	1		

	4	5	
		2	7
3	1		

```
\Tableau{ 45,2751,31 }
\SkewTableau{1,2}{ 45,27,31 }
```

2A.50

Even though it is possible to draw compositions, and tableaux of composition shape, compositions are not supported in the sense that many of the options/keys assume that the diagrams are of partition shape. In addition, some options assume that tableaux and diagrams do not contain empty rows, and that the diagrams are *connected*. (For disconnected tableaux and diagrams, see the `\Multidiagram` and `\Multitableau` commands in [Section 2G](#).)

2A.5. *Drawing order*. When `TikZ` constructs a picture, later objects are drawn over the top of earlier ones, which can obscure earlier features. Consequently, for more complicated diagrams and tableaux it helps to know the order in which the different parts of these diagrams are drawn, which is the following:

- First, the tableau entries that use the default styling are placed, in the order that they appear in the *tableau specifications*.
- Secondly, the styled tableau entries are placed, in the order that they appear in the *tableau specifications*.
- Next, the nodes in any *ribbons* are placed, again in the order that the ribbons are listed. For each ribbon, first the border of the ribbon is drawn, together with any styling for the ribbon, and then the boxes in the ribbon, together with any styling, are added.
- When enabled by *skew border*, the skew border is drawn, after which the border of the tableau is drawn.
- The *dotted rows* and *dotted columns* keys are applied to replace the specified rows and columns with dots.
- Finally, any *snoobs* are drawn, in the order that they are listed.

All of the tableau and Young diagram commands use this drawing order.

2A.6. *Special characters*. The characters `[]`, `,`, and `*` have special meanings in the *tableau specifications*. Enclose these characters in braces when you want to use these characters as entries in a tableau:

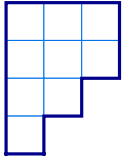
1	3	,	5	[]
8	[			
*				

```
\Tableau{ 13{,}5[], 8{[}, {{*}} }
```

2A.51

There is no need to enclose `[]` in braces because it only becomes special when it is preceded by a matching `[` character.

2B. Young diagrams. A *Young diagram*, or simply a *diagram*, is an unlabelled tableau. Diagrams can be drawn using the `\Tableau` command:



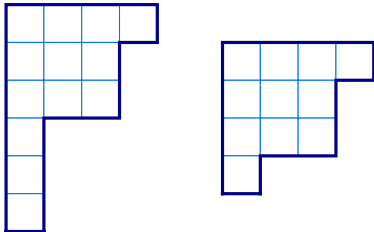
```
\Tableau{~~~,~~~,~~,~}
```

2B.1

This approach works, but it is cumbersome, hard to proofread, and easy to miscount. For this reason, `aTableau` provides the `\Diagram` command. The `\Diagram` command uses almost the same syntax as the `\Tableau` command:

```
\Diagram (x,y) [options] {partition}
```

As with the `\Tableau` command, the (x,y) -coordinates are needed if, and only if, the diagram is inside a `tikzpicture` environment. The `\Diagram` command allows the *partition* to be specified as either a comma-separated list of *weakly decreasing positive integers*, or by using *exponential notation*:



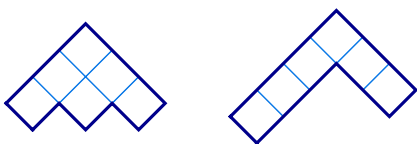
```
\Diagram{4,3^2,1^3}  
\Diagram{4,3,3,1}
```

2B.2

Internally, `\Diagram` actually does something like the first example in this section, so all of the options for the `\Tableau` command can be used with `\Diagram`. Rather than repeating all of the `\Tableau` options in this section, we highlight the more interesting options together with some new, diagram specific, options.

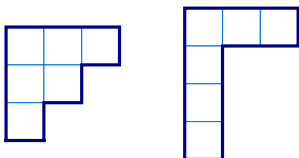
Just like the `\Tableau` command, `\Diagram` supports the four different conventions for diagrams, with `english` being the default:

english
french
ukrainian
australian (default: english)



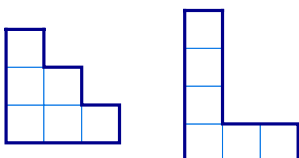
```
\Diagram[australian]{3,2,1}  
\Diagram[australian]{3,1^3}
```

2B.3



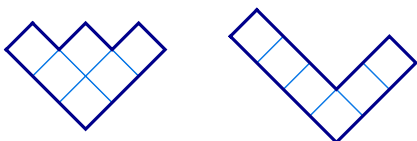
```
% English is the default convention  
\Diagram[english]{3,2,1}  
\Diagram{3,1^3}
```

2B.4



```
\Diagram[french]{3,2,1}  
\Diagram[french]{3,1^3}
```

2B.5



```
\Diagram[ukrainian]{3,2,1}  
\Diagram[ukrainian]{3,1^3}
```

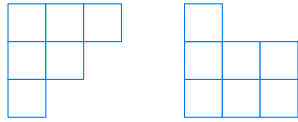
2B.6

`border` (default: `true`)
`no border` (default: `false`)

[accepts: `true/false`]
[accepts: `true/false`]

aTableau

If `border` is true, which it is by default, then the border of the tableau is drawn. If `border` is false, then the border of the tableau is not drawn. Notice that the border refers only to the outside border of the diagram, and not to the internal borders of the boxes on the diagram, which can be disabled using `border only`.



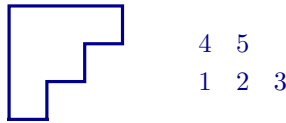
```
\Diagram[border=false]{3,2,1}
\Diagram[no border, french]{3^2,1}
```

2B.7

`border only` (default: `false`)

[accepts: false/true]

When `border only` is set to true the internal walls around boxes are not drawn.



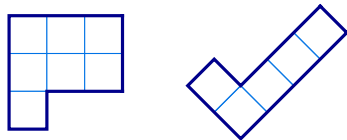
```
\Diagram[border only]{3,2,1}
\Tableau[french, border only, no border]{123,45}
```

2B.8

`conjugate` (default: `false`)

[accepts: false/true]

Draws the conjugate diagram, which has the rows and columns interchanged.



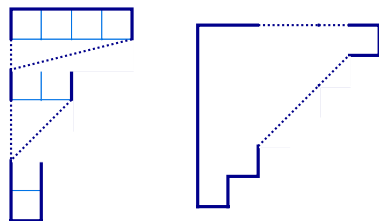
```
\Diagram[conjugate]{3,2,2}
\Diagram[ukrainian, conjugate]{2,1^3}
```

2B.9

`dotted columns`
`dotted rows`

[accepts: comma separated list of columns]
[accepts: comma separated list of rows]

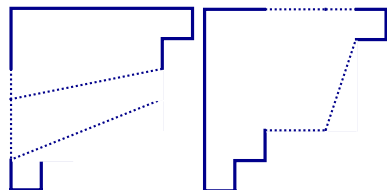
The `dotted rows` and `dotted columns` keys can be used to draw diagrams where the specified rows and columns are replaced with dots. This makes it possible to draw *generic* diagrams, where the number of rows and columns is not fully specified. Instead of `dotted columns`, you can use the equivalent short-hand key `dotted cols`.



```
\aTabset{scale=0.8}
\Diagram[dotted rows={2,4,5}]{4^2,2^2,1^3}
\Diagram[dotted columns={5,3,4},
border only]{6,5,4,3,2,1}
```

2B.10

As shown above, when row and columns indices are in increasing *consecutive* order, then the corresponding rows or columns are replaced as a block. Non-consecutive indices are treated separately.

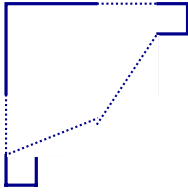


```
\aTabset{scale=0.8, border only}
\Diagram[dotted rows={4,5,3}]{6,5,5,5,2,1}
\Diagram[dotted cols={5,3,4}]{6,5,5,5,2,1}
```

2B.11

The `dotted rows` and `dotted columns` keys work by overwriting the content on the specified rows and columns, so anything that was drawn in these rows and columns will disappear. The `dotted rows` and `dotted columns` keys can be used with the `\Tableau` command, with the caveat that this will remove all boxes, with their contents, that are in the specified rows and columns.

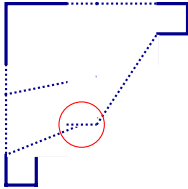
When the `dotted rows` and `dotted columns` keys are used together, the `dotted rows` are applied first.



```
\Diagram[scale=0.8, border only,
  dotted rows={4,5},
  dotted cols={4,5},
] {6,5,5,5,2,1}
```

2B.12

When using these two keys together, you need to be careful when choosing which rows and columns are removed because, otherwise, artifacts can remain. Ideally, the endpoints of the rows and columns being removed should not overlap. For example, the circled horizontal dots below are a “user error”, not a bug, because these dots come from trying to use `dotted rows` to remove row three:



```
\Diagram[scale=0.8, border only,
  dotted rows={4,5,3}, dotted cols={4,5,3},
  tikz after = {\draw[red] (1,-1.6) circle(0.3);}
] {6,5,5,5,2,1}
```

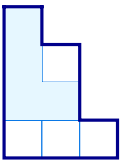
2B.13

The `dotted rows` and `dotted columns` keys are not designed to be used with the `conjugate key`.

`ribbons`
`snobs`

[accepts: ribbon specification]
[accepts: ribbon specification]

Use `ribbons` and `snobs` to add ribbons to a diagram.



```
\Diagram[french, ribbons={*22*c*r*r}] {3,2^2,1}
```

2B.14

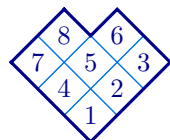
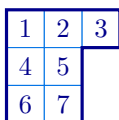
For more details about the ribbon specifications, with examples, see [Section 2F](#), which describes the `\RibbonTableau` command.

`entries`

[accepts: contents,first,hooks,last,residue]

2B.1. *Diagrams with entries.* The `entries` option provides a shortcut for drawing some common tableaux of a specified shape. The possible choices are:

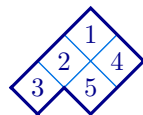
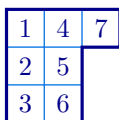
- `entries=first` print the *first standard* tableau of this shape, with respect to the Bruhat order. This is the tableau that has the numbers $1, 2, \dots, n$ entered in order from top left to right along down successive rows (with appropriate modifications for the non-English conventions).



```
\Diagram[entries=first] {3,2,2}
\Diagram[ukrainian, entries=first] {3^2,2}
```

2B.15

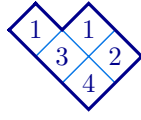
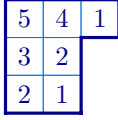
- `entries=last` print the *last standard* tableau of this shape, with respect to the Bruhat order. This is the tableau that has the numbers $1, 2, \dots, n$ entered in order from top to bottom down successive columns (with appropriate modifications for the non-English conventions).



```
\Diagram[entries=last] {3,2,2}
\Diagram[australian, entries=last] {2^2,1}
```

2B.16

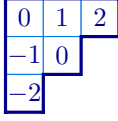
- `entries=hooks` print the diagram where each box is labelled by the corresponding *hook length*. If λ is a partition and λ' is its conjugate partition then the hook length of the box in row i and column j is $\lambda_i - i + \lambda'_j - j + 1$.



```
\Diagram[entries=hooks]{3,2,2}
\Diagram[ukrainian,entries=hooks]{2^2,1}
```

2B.17

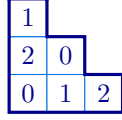
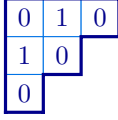
- `entries=contents` print the diagram where each box is labelled by its *content*, which is the row index minus the column index.



```
\Diagram[entries=contents]{3,2,1}
\Diagram[french,entries=contents]{2^2,1}
```

2B.18

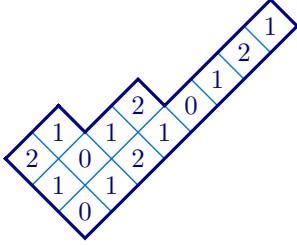
- `entries=residues` print the diagram where each box is labelled by its *residue*, which is the row index minus the column index modulo an integer $e \geq 2$, which must also be supplied.



```
\Diagram[entries=residues, e=2]{3,2,1}
\Diagram[french, entries=residues, e=3]{3,2,1}
```

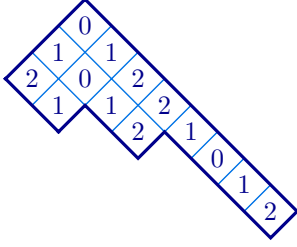
2B.19

By default, residue sequences for affine type $A_e^{(1)}$, or the symmetric group and friends, are given. In addition, residues in affine types $C_e^{(1)}$, $A_{2e}^{(2)}$ and $D_e^{(2)}$, which can be accessed using `type=C`, `type=AA`, and `type=DD`, respectively.



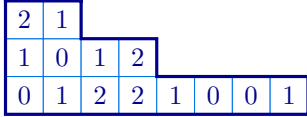
```
% affine type C
\Diagram[ukrainian,entries=residues, e=2, type=C]
{8,4,2}
```

2B.20



```
% twisted affine type A
\Diagram[australian, entries=residues, e=2,
type=AA]
{8,4,2}
```

2B.21



```
% twisted affine type D
\Diagram[french, entries=residues, e=2, type=DD]
{8,4,2}
```

2B.22

2C. Skew tableaux and skew diagrams. A partition μ is *contained* in another partition λ , which is written as $\mu \subseteq \lambda$, if $\mu_k \leq \lambda_k$, for $k \geq 0$. If $\mu \subseteq \lambda$ then the *skew partition* $\lambda/\mu = \{(r, c) \in \lambda \mid (r, c) \notin \mu\}$ is the set of nodes that are in λ and not in μ . In this manual, μ is the *inner shape* and λ/μ is the outer shape. A *skew tableau* is a labelling of the nodes in the diagram of a skew partition. Skew partitions and skew tableau can be drawn using the commands:

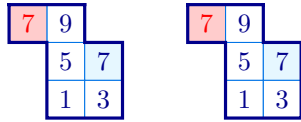
```
\SkewDiagram (x,y) [options] {inner shape} {outer shape}
\SkewTableau (x,y) [options] {inner shape} {skew tableau specifications}
```

As with the previous commands, the (x, y) -coordinates should be given if, and only if, the picture is inside a `tikzpicture` environment. The inner and outer shapes can either be given as a comma-separated list of non-negative integers, or in exponential notation.

Skew tableau and skew diagrams should always be connected. Use `\Multidiagram` and `\Multitableau` to draw disconnected diagrams.

In fact, the `\SkewTableau` is just an alias for the `\Tableau`, using the `skew` key to set the inner shape. For this reason, all of the options for the `\Tableau` command can be used with `\SkewTableau`. The entries in a `\SkewTableau` are specified in exactly the same way as in the `\Tableau` command. In

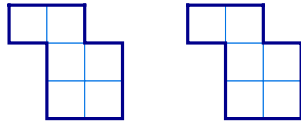
particular, the entries of skew tableaux can, optionally, be given style prefixes, both using a `*` to add the current `star style`, or arbitrary `TikZ`-styling using `[...]`.



```
\tikzset{R/.style={fill=red!20,text=red}}
\SkewTableau[french]{1^2}{13,5*7,[R]79}
\Tableau[french,skew={1^2}]{13,5*7,[R]79}
```

2C.1

Similarly, the `\SkewDiagram` command is an alias for the `\Diagram` command, with a `skew` key.



```
\SkewDiagram[french]{1^2}{2^3}
\Diagram[french,skew={1^2}]{2^3}
```

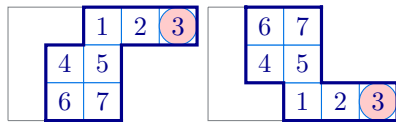
2C.2

Since the `\SkewTableau` and `\SkewTableaua` commands are shortcuts, all of the options for the `\Tableau` and `\Diagram` commands can be used with the `\SkewTableau` and `\SkewDiagram` commands, respectively. In addition, the following options are supported:

`skew border` (default: `false`)
`no skew border` (default: `true`)

[accepts: `true/false`]
 [accepts: `false/true`]

These two keys are inverse to each other. When `skew border` is `true`, the border of the (inner) skew shape is drawn.



```
\tikzset{R/.style={fill=red!20,circle}}
\SkewTableau[skew border]{2,1^2}{12[R]3,45,67}
\SkewTableau[french,skew border]{2,1^2}{12[R]3,45,67}
```

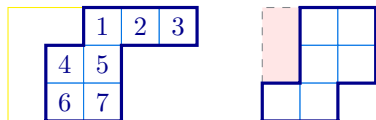
2C.3

Note that `skew border` only draws the border of the skew shape and not the boxes inside the inner skew shape. Use `skew boxes` to draw the interior walls of the boxes in the skew shape.

`skew border style` (default: `draw=` , `fill=` , `very thick`)

[accepts: `TikZ`-styling]

Use `skew border style` to change the style of the skew border. Since the skew border is only drawn when `skew border` is set, the `skew border style` does not have any effect unless `skew border` has been set to `true`.



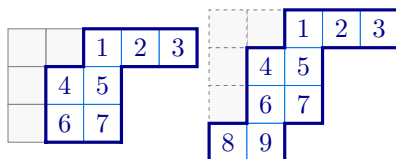
```
\SkewTableau[skew border style={draw=yellow},
  skew border]{2,1^2}{123,45,67}
\SkewDiagram[skew border style={dashed,fill=red!10},
  skew border]{1^2}{2^3}
```

2C.4

`skew boxes` (default: `false`)
`skew box style` (default: `thin,gray`)

[accepts: `true/false`]
 [accepts: `TikZ`-styling]

The `skew boxes` key draws the inner walls in the inside the inner partition of a skew shape. Use `skew box style` to change the default shading of these boxes.



```
\SkewTableau[skew boxes]{2,1^2}{123,45,67}
\SkewTableau[skew boxes,
  skew box style={dash pattern=on 1pt off 2pt}
]{2,1^2}{123,45,67,89}
```

2C.5

Using the options `skew border` and the `skew boxes` together gives the skew (inner) shape both a inner and outer borders.



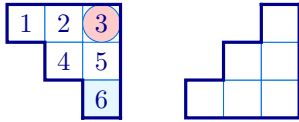
```
\SkewTableau[skew border, skew boxes]{2^2,1}{123,45,67}
```

2C.6

2D. Shifted tableaux and shifted diagrams. *Strict partitions*, which have strictly increasing parts, and *shifted tableaux*, which are of strict partition shape, appear in several places in representation theory, such as the projective representations of the symmetric groups. These tableaux and diagrams are drawn with row r shifted by $(r - 1)$ -units along the row, so that the first box in each row has content 0. These diagrams and tableaux can be drawn using the following commands:

```
\ShiftedDiagram (x,y) [options] {partition }
\ShiftedTableau (x,y) [options] {tableau specification}
```

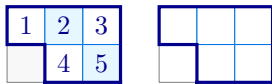
As usual, the (x, y) -coordinates are necessary if, and only if, these commands are used inside a `tikzpicture` environment. The partition in a `\ShiftedDiagram` can be given using exponential notation and the tableau specification for a `\ShiftedTableau` can include the usual optional style prefixes.



```
\tikzset{R/.style={fill=red!20,circle}}
\ShiftedTableau{12[R]3,45,*6}
\ShiftedDiagram[french]{3,2,1}
```

2D.1

In the literature, shifted tableaux and shifted diagrams almost always have strict partition shape, however, this is not enforced by these commands. Under the hood, the `\ShiftedTableau` is the `\Tableau` command with the `shifted` option set to true. Similarly, `\ShiftedDiagram` is the same as using the `\Diagram` command with the `shifted` option. Consequently, all of the options for the `\Tableau` and `\Diagram` commands can be used with `\ShiftedTableau` and `\ShiftedDiagram`, respectively. In particular, options like `skew boxes` can be used to highlight the shifted part of the diagram



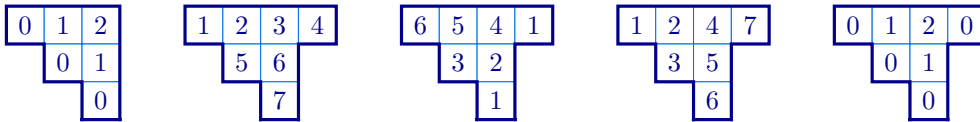
```
\ShiftedTableau[skew boxes]{1*23,4*5}
\ShiftedDiagram[skew border]{3,2}
```

2D.2

The `entries` key works as expected for shifted diagrams:

```
\ShiftedDiagram[entries=contents]{3,2,1}
\ShiftedDiagram[entries=first]{4,2,1}
\ShiftedDiagram[entries=hooks]{4,2,1}
\ShiftedDiagram[entries=last]{4,2,1}
\ShiftedDiagram[entries=residues,e=3]{4,2,1}
```

2D.3

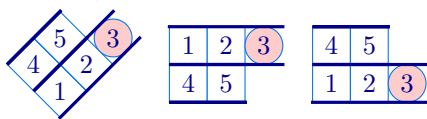


The definition of hook lengths for shifted diagrams can be found in the literature!

2E. Tabloids. A *tabloid* is an equivalence class of tableau, where two tableaux are equivalent if they have the same entries in each row, up to a permutation. In the literature, tabloids are usually drawn with lines above and below each row, and without side borders. Tabloids can be drawn with the `\Tabloid` command.

```
\Tabloid (x,y) [options] {partition }
```

The `\Tabloid` command functions in the same way as the `\Tableau` command, except that it draws tabloids. In fact, the `\Tabloid` is a special case of the `\Tableau` command with the `tabloid` option set.

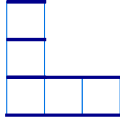
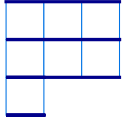


```
\tikzset{R/.style={fill=red!20,circle,minimum size=5mm}}
\Tabloid[ukrainian]{12[R]3,45}
\Tabloid[english]{12[R]3,45}
\Tabloid[french]{12[R]3,45}
```

2E.1

The `aTableau` package does not provide a dedicated command for tabloid diagrams, however, they can be drawn using the `\Diagram` command together with the `tabloid` option:

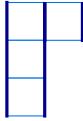
aTableau



```
\Diagram[tabloid]{3^2,1}
\Diagram[tabloid,french]{3,1^2}
```

2E.2

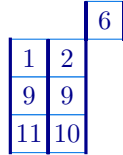
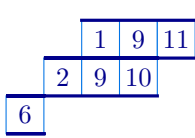
Some authors work with *column tabloids*. Such tableaux do not have a dedicated `aTableau` command, however, they can be drawn using the `conjugate` option:



```
\Tabloid[conjugate]{123,4}
\Diagram[conjugate,tabloid]{3,1}
```

2E.3

Similarly, skew tabloids can be drawn using the `skew` option.



```
\Tabloid[skew={2,1}]{19{11},29{10},6}
\Tabloid[conjugate,skew={1^2}]{19{11},29{10},6}
```

2E.4

2F. Ribbon tableaux. A *ribbon* in a tableau, or a diagram, is a connected strip of boxes R that are totally ordered by their contents. (Recall from ?? that the content of the box in row a and column b is $b - a$.) Therefore, a ribbon does not contain a $\boxplus$ -square, and if $(a, b) \in R$ then at most one of $(a + 1, b)$ and $(a, b - 1)$ belongs to R . The *head* of a ribbon R is the unique node of maximal content. A *ribbon tableau* is a tableau that is tiled by ribbons. As a box is a ribbon of length 1, so every tableau is a ribbon tableau.

`\RibbonTableau (x,y) [options] {ribbon specifications}`

Like the other commands, the (x, y) -coordinates are optional and are required if, and only if, the diagram is being used as part of a `tikzpicture` environment. All options for the `\Tableau` command can be used for the ribbon command, so we refer to [Section 2A](#) for a description of the available options.

2F.1. Ribbon specifications. Unlike the `\Tableau` command, the entries of a ribbon tableau are given by ribbon specifications (rather than tableau) specifications. Ribbon specifications are also used by the `ribbons` and `snobs` options discussed in [Subsection 2A.3](#).

To understand the ribbon specifications, observe that if (a, b) is the head of a ribbon, then we can walk along the ribbon by specifying whether the row index increases or the column index decreases. That is, a ribbon is uniquely determined by specifying its head (a, b) together with a sequence of r 's and c 's to indicate when the row index increases, or the column index decreases, respectively. For example, vertical and horizontal ribbons are given by a sequences of r 's and c 's, respectively:

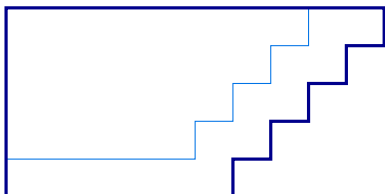


```
\aTabset{align=top, no border}
\RibbonTableau{11rrrr}
\RibbonTableau{16ccccc}
```

2F.1

The border of a ribbon tableau is the smallest (skew) partition that contains all of its ribbons. As the last example shows, the `no border` key disables drawing of the ribbon tableau border because, otherwise, we would not be able to see the ribbon border in these examples since the ribbon is the full diagram in these two cases.

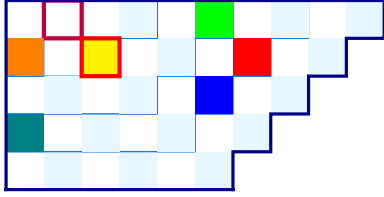
The following diagram shows a see-sawing ribbon with head in row 1 and column 10, and tail in row 5 and column 1.



```
\RibbonTableau{1{10}crrcrrccccc}
```

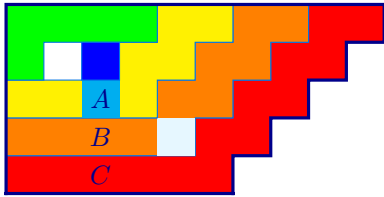
2F.2

As with tableau entries, each box in a ribbon can be preceded with a style specification, either using `*` for the [star style](#) or using `[...]` for more customised style specifications.



```
\RibbonTableau{
  *1{10}c*rc*rc*rc*rc*cc*cc,
  *18c[fill=red]rc[fill=blue]rc*rc*cc[fill=teal]c,
  [fill=green]16c*rc*rc*cc,
  *14c[{draw=purple,ultra thick}]cc[fill=orange]r,
  [{fill=yellow,draw=red,ultra thick}]23
}
```

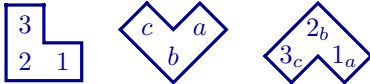
In particular, the yellow box in this example highlights that boxes are ribbons of length 1. Most of the time, you want to style the entire ribbon rather than styling the individual boxes in the ribbon. Optionally, the style for the entire ribbon can be given inside parentheses (...) at the *start* of the ribbon. Unlike the other style specifications, the `*`-shorthand for the [star style](#) cannot be applied to an entire ribbon.



```
\RibbonTableau{
  (fill=red)1{10}crcrcrcrccc_Ccc,
  (fill=orange)18crcrc*rcc_Bcc,
  (fill=yellow)16crcr[fill=cyan]c_Acc,
  (fill=green)14cccr,
  (fill=blue)23
}
```

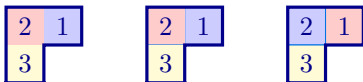
Any styles for a ribbon are applied to the region bounded by the closed path given by the border of the ribbon. The ribbon style is applied first, after which any styled boxes in ribbon are drawn. As this example shows, the optional style of any box in the ribbon takes precedence over the style of the full ribbon.

Finally, it is often necessary to label some of the boxes in the ribbon. Optionally, text can be added to the boxes in a ribbon by supplying it as a *subscript* to the corresponding entry in the ribbon specification:



```
\RibbonTableau[french] { 12_1 c_2 r_3 }
\RibbonTableau[ukrainian] { 12_a c_b r_c }
\RibbonTableau[australian]{ 12_{1_a} c_{2_b} r_{3_c} }
```

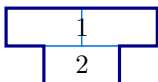
Here we have put spaces between the entries for the different boxes in the ribbon for clarity. There are often many ways to draw the same ribbon tableau, especially if the entries have styling.



```
\tikzset{Fill/.style={fill=#1!20!white},
  B/.style={Fill=blue}, R/.style={Fill=red},
  Y/.style={Fill=yellow}}
\RibbonTableau{ [B]12_1 [R]c_2 [Y]r_3 }
\RibbonTableau{ [B]12_1, [R]11_2, [Y]21_3 }
\Tableau{ [B]2 [R]1, [Y]3 }
```

As shown, the easiest way to draw this particular tableau is using the `\Tableau` command.

There is no dedicated syntax for putting a label on the boundary between two boxes in the diagram, but this can be done using the named anchors of (1.1).



```
\RibbonTableau[shifted, tikz after={
  \node at (B-1-2.east){$1$};
  \node at (B-2-2.east){$2$};
}]{ 14c, 12c, 23c }
```

We have now seen the full ribbon specifications. To summarise:

- The ribbon specification starts with optional [TikZ](#)-styling for the ribbon, enclosed in (...).
- The first entry in the ribbon is given as `ab`, if the head of the ribbon is in row `a` and column `b`. The remaining entries in the ribbon are specified by a sequence of `r`'s and `c`'s, depending on whether

the next box in the ribbon is given by increasing the row index, or decreasing the column index, respectively.

- Optionally, each box in the ribbon, including the head, can be given **TikZ**-styling by prefixing it with a *, which gives the ribbon **star style**, or by giving **TikZ**-styling keys inside square brackets [...].

Finally, when drawing the border, ribbons in a `\RibbonTableau` are assumed to be contained in the smallest Young diagram that contains the ribbons. This said, it is up to you to ensure that the boxes in the ribbon not have negative row or column indices. Inside a `\Diagram` or `\Tableau` command, any ribbons added using `ribbons`, or `snobs`, do not affect the border of the diagram or tableau, and it is your responsibility to ensure that the boxes in the ribbon are inside the diagram or tableau. This said, it is a useful feature to be able to place boxes outside of the diagrams, which can be done using `ribbons` or `snobs` as these do not contribute to the border of a ribbon tableaux.

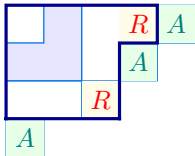
ribbons

[accepts: comma separated ribbon specifications]

snobs

[accepts: comma separated ribbon specifications]

You can add `ribbons` and `snoBs` to a `\RibbonTableau` in exactly the same way that you add them to `\Tableau`. At first sight, this seems unnecessary because ribbon tableaux are composed of ribbons, however, the point is that any ribbon added to a tableau using `ribbons` or `snoBs` is not assumed to be inside the border of the tableau.



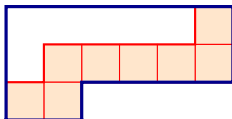
```
\tikzset{A/.style={fill=green!10, text=teal},
R/.style={fill=yellow!10, text=red}}
\RibbonTableau[
  ribbons={ [A]15_A, [A]24_A, [A]41_A, [R]14_R, [R]33_R}
  {11, (fill=blue!10)12rc, 14crrcc}
```

ribbon style (default: draw=, thin)
ribbon box

TikZ-styling

TikZ-styling

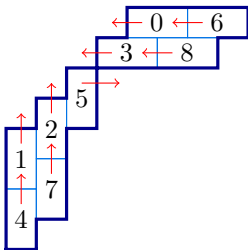
Use the `ribbon style` and `ribbon box` options to append to the default styling of the ribbons and the boxes in the ribbons, respectively. By default, ribbons have an inner wall and the boxes in ribbons have no styling.



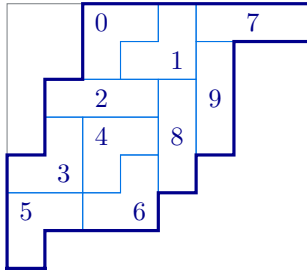
```
\RibbonTableau[ribbon style={draw=red, thick},
  ribbon box={fill=orange!20, draw=red}]{16rccccrc}
```

2F.9

2F.2. *Some ribbed examples.* We close this section showing how to draw some tableaux that appear in the literature.

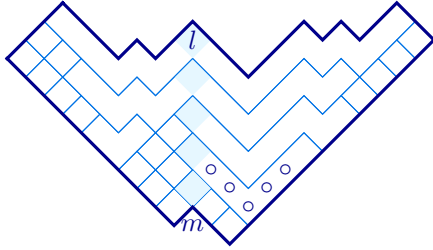


```
\RibbonTableau[,skew={4,3,2,1},scale=0.8,
tikz after = {
    % node labels
    \foreach \r/\c/\l in {1/6/0, 1/8/6, 2/5/3, 2/7/8}
        { \node at (B-\r-\c.west){\l};}
    \foreach \r/\c/\l in {5/1/1, 4/2/2, 3/3/5, 7/1/4, 6/2/7}
        { \node at (B-\r-\c.south){\l};}
    % arrows
    \foreach \a/\b in {1/5, 1/7, 2/4, 2/6}
        { \draw[red,->](B-\a-\b.center)---+(-0.4,0); }
    \draw[red,->](B-3-3.center)---+(0.5,0);
    \foreach \a/\b in {5/1, 7/1, 4/2, 6/2}
        { \draw[red,->](B-\a-\b.center)---+(0,0.4); }
}
]{
16c,18c, 25c, 27c, 33r, 42r, 51r, 62r,71r
}
```



```
\RibbonTableau[skew={2^2,1^2},skew border]{
  14c_0r,15r_1c, 18c_7c,26r_9r,
  34c_2c,35r_8r,42r_3c, 44c_4r,
  54r_6c,62c_5r
}
```

2F.11



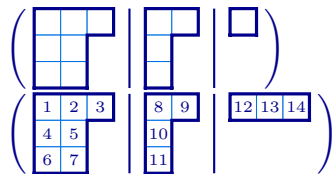
```
\RibbonTableau[skew={1^2}, ukrainian, scale=0.7,
  ribbons={(draw=none)21_m}
]{
  12,16c_\circ c_\circ c_\circ r_\circ r_\circ r_\circ *,
  18crrrrr*r,19,1{10},1{11},1{12},
  22,29crrrrr*rccrrr,
  2{12}ccrrrrrr*r_lccrrrr,
  31,*32,41,42,51,52,53,
  62cr,81,91,{10}1,{10}2,{11}1,{11}2
}
```

2F.12

2G. Multidiagrams and multitableaux. Multitableaux and multidiagrams are ℓ -tuples of tableaux and diagrams, respectively. These diagrams appear in many places in representation theory, such as when considered representations of wreath products and cyclotomic Hecke algebras. For convenience, **aTableau** provides the `\Multidiagram` and `\Multitableau` commands for drawing these diagrams, even though it is not difficult to construct such tableaux and diagrams using the `\Tableau` and `\Diagram` commands.

```
\Multidiagram (x,y) [options] {multipartition specifications}
\Multitableau (x,y) [options] {multitableau specifications}
```

Our preferred notation for a multipartition λ is to write $\lambda = (\lambda^{(1)}|\lambda^{(2)}|\dots|\lambda^{(\ell)})$, where $\ell \geq 1$ is the *level* and the partitions $\lambda^{(1)}, \dots, \lambda^{(\ell)}$ are the *components* of λ . Similarly, a multitableau is written as $t = (t^{(1)}|\dots|t^{(\ell)})$, where the tableaux $t^{(1)}, \dots, t^{(\ell)}$ are the *components* of t . Accordingly, in the `\Multidiagram` and `\Multitableau` commands, the pipe symbol `|` is used to separate components:



```
\aTabset{scale=0.7}
\Multidiagram{3,2^2|2,1,1|1}
\Multitableau[box font=\tiny]
{ 123,45,67|89,{10},{11}}|{12}{13}{14} }
```

2G.1

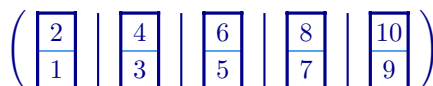
As shown, the `\Multidiagram` command accepts exponential notation for partitions. The entries in a `\Multitableau` can be given optional styling specifications, exactly as inside the `\Tableau` command.



```
\tikzset{R/.style={fill=red!10,text=magenta}}
\Multitableau{1*24,*68|[R]9{10},{11}}
```

2G.2

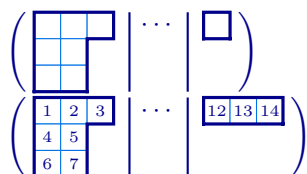
Multitableaux and multidiagrams can, in principle, have arbitrary level. In practice, the level is constrained by the page width.



```
\Multitableau[french]{1,2|3,4|5,6|7,8|9,{10}}
```

2G.3

Both the `\Multitableau` and `\Multidiagram` commands accept a special component, `\dots`, that is used to add dots between the separators:



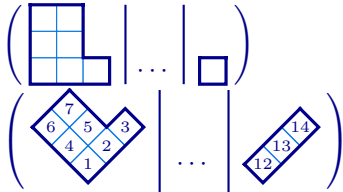
```
\aTabset{scale=0.7}
\Multidiagram{3,2^2|\dots|1}
\Multitableau[box font=\tiny]
{123,45,67|\dots|{12}{13}{14}}
```

2G.4

The `\Multitableau` and `\Multidiagram` commands accept most of the options of the `\Tableau` and `\Diagram` commands, together with a few options that are specific to multitableaux and multidiagrams. This section describes the new options, and highlights how some of the other options are used.

english
french
ukrainian
australian (default: english)

All four tableaux conventions are supported for multitableaux and multidiagrams, with the caveat that all component diagrams use the same convention.

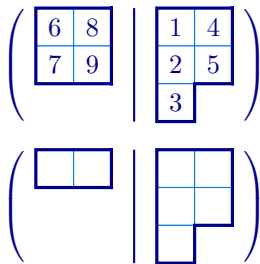


```
\aTabset{scale=0.7}
\Multidiagram[french]{3,2^2|\dots|1}
\Multitableau[ukrainian, box font=\tiny]
{123,45,67|\dots|\{12\}\{13\}\{14\}}
```

2G.5

conjugate

The `conjugate` option prints the conjugate multitableaux and multidiagrams. Conjugation for multitableaux and multidiagrams reverses the order of the components and then conjugates the component diagrams.



```
\Multitableau[conjugate]{ 123,45 | 67,89}
\Multidiagram[conjugate]{ 3,2 | 1^2 }
```

2G.6

delimiters (default: ())
left delimiter (default: ())
right delimiter (default: ())

[accepts: pair of L^AT_EX delimiters]
[accepts: a L^AT_EX delimiter]
[accepts: a L^AT_EX delimiter]

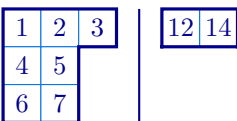
The delimiter options control the delimiters that are put on the left and right of multitableaux and multidiagrams. These options accept any L^AT_EX delimiter.



```
\aTabset{scale=0.7}
\Multidiagram[left delimiter=\langle,
right delimiter={}] {2,1|1}
\Multidiagram[delimiters={\|}{\}] ]{1^2|2}
```

2G.7

To remove a delimiter, set `left delimiter` or `right delimiter` to empty.



```
\Multitableau[left delimiter=, right delimiter=]
{123,45,67|\{12\}\{14\}}
```

2G.8

empty (default: `\textendash`)

[accepts: any L^AT_EX]

The `empty` option determines what is printed when a component diagram is empty. By default, the en-dash symbol – is used.

$$\left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 6 & 7 \\ \hline 8 & 9 \\ \hline \end{array} \right)$$

$$\left(\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \right) \emptyset \left(\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \right)$$

```
\Multitableau{ 123,45 | | 67,89}
\Multidiagram[empty=${\emptyset}]{ 3,2 | | 2^2 }
```

2G.9

entries [accepts: contents,first,hooks,last,residues] **charge** [accepts: |-separated list of charges]

Use the **entries** key to draw particular types of tableau of the specified shape. Use **charge** to provide offsets in each component when using **entries=contents** and **entries=residues** (and no charge should be set when using **entries=first** and **entries=last**).

The possible choices are:

- **entries=first** print the *first standard* tableau of this shape, which has the numbers $1, 2, \dots, n$ entered in order from top left to right along down successive rows (with appropriate modifications for the non-English conventions).

$$\left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline 5 & & \\ \hline \end{array} \right) \left(\begin{array}{|c|c|} \hline 6 & 7 \\ \hline 8 & 9 \\ \hline 10 & \\ \hline \end{array} \right)$$

```
\Multidiagram[entries=first]{3,1^2|2^2,1}
```

2G.10

- **entries=last** print the *last standard* tableau of this shape, which has the numbers $1, 2, \dots, n$ entered in order from top to bottom down successive columns, and left to right along components (with appropriate modifications for the non-English conventions).

$$\left(\begin{array}{|c|c|c|} \hline 5 & 8 & 11 \\ \hline 6 & 9 & \\ \hline 7 & 10 & \\ \hline \end{array} \right) \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline & \\ \hline \end{array} \right)$$

```
\Multidiagram[entries=last]{3,2,2|2^2}
```

2G.11

- **entries=hooks** print the diagram where each box is labelled by the corresponding *hook length*. If λ is a partition and λ' is its conjugate partition then the hook length of the box in row i and column j is $\lambda_i - i + \lambda'_j - j + 1$.

$$\left(\begin{array}{|c|c|c|} \hline 5 & 2 & 1 \\ \hline 2 & & \\ \hline 1 & & \\ \hline \end{array} \right) \left(\begin{array}{|c|c|c|} \hline 5 & 4 & 3 \\ \hline 4 & 3 & 2 \\ \hline 3 & 2 & 1 \\ \hline \end{array} \right)$$

```
\Multidiagram[entries=hooks]{3,1^2|3^3}
```

2G.12

- **entries=contents** print the diagram where each box is labelled by its *content*, which is the row index minus the column index.

$$\left(\begin{array}{|c|c|c|} \hline 0 & 1 & 2 \\ \hline -1 & 0 & \\ \hline -2 & & \\ \hline \end{array} \right) \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 2 \\ \hline & \\ \hline \end{array} \right)$$

```
\Multidiagram[entries=contents,
charge={0,2}]{3,2,1|2^2}
```

2G.13

- **entries=residues** print the diagram where each box is labelled by its *residue*, which is the row index minus the column index modulo some integer e , which must also be supplied.

$$\left(\begin{array}{|c|c|c|} \hline 0 & 1 & 0 \\ \hline 1 & 0 & \\ \hline 0 & & \\ \hline \end{array} \right) \left(\begin{array}{|c|} \hline 0 \\ \hline 1 \\ \hline 0 \\ \hline \end{array} \right)$$

```
\Multidiagram[entries=residues, e=2]{3,2,1|1^3}
```

2G.14

$$\left(\begin{array}{|c|c|c|} \hline 2 & & \\ \hline 0 & 1 & \\ \hline 1 & 2 & 0 \\ \hline \end{array} \right) \left(\begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline \\ \hline \end{array} \right)$$

```
\Multidiagram[french,entries=residues,
e=3,charge={1,2}]{3,2,1|1^2}
```

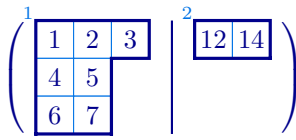
2G.15

As in [Subsection 2B.1](#), residues for the other affine Cartan types can be specified using **type=??**.

label

[accepts: a |-separated list of labels]

Use the **label** key to add labels to the component diagrams.



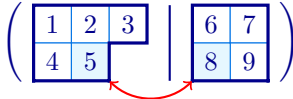
```
\Multitableau[label={1|2}]{123,45,67|{12}{14}}
```

2G.16

name (default: **B**)

[accepts: character]

The **name** option is used to change the default prefix for the box node names. In a tableau, or diagram, the box names depend on the *row* and *column* indices. In a multitableau, or multidiagram, the box names depend on the *component*, *row* and *column* indices. Explicitly, by default, in multitableaux and multidiagrams the names of the take the form B-k-r-c, where k is the component index, r is the row index, and c is the column index of the box. The B can be changed to anything that you like using the **name** key.



```
\Multitableau[name=A, tikz after = {
  \draw[red, thick,<->](A-1-2-2.south east) to
    [out=315,in=225] (A-2-2-1.south west);
}] { 123,4*5 | 67,*89}
```

2G.17

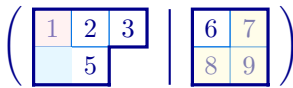
ribbons

[accepts: a |-separated list of ribbon specifications]

snobs

[accepts: a |-separated list of ribbon specifications]

The **ribbons** and **snobs** keys work in exactly the same way for multitableaux and multidiagrams as they do for tableaux and diagrams except that the ribbons for different components are separated by a pipe |.



```
\tikzset{R/.style={fill=red!10, opacity=0.5},
  Y/.style={fill=yellow!20, opacity=0.5}}
\Multitableau[ribbons={ (R) 11*r | (Y) 12rc}]
{ 123,45 | 67,89}
```

2G.18

rows

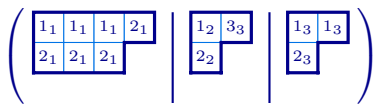
[accepts: a decimal number]

The **rows** option sets the number of rows covered by the delimiters in multitableaux and multidiagram. By default, the `\Multitableau` and `\Multidiagram` commands choose the size of the delimiters to match the component diagrams, which may not be what you want especially when using the **australia** and **ukrainian** conventions. The value of **rows** can be a decimal number.



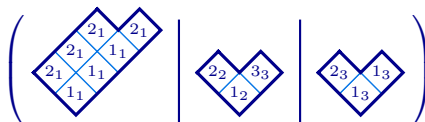
```
\Multidiagram[ukrainian,scale=0.5]{3,2^2|...|2,1^2}
```

2G.19



```
\Multitableau[rows=3,scale=0.8,box font=\tiny]
{{1_1}{1_1}{1_1}{2_1},{2_1}{2_1}{2_1} |
{1_2}{3_3},{2_2}} |
{1_3}{1_3},{2_3}}
```

2G.20



```
\Multitableau[ukrainian,rows=2.5,scale=0.8,box font=\tiny]
{{1_1}{1_1}{1_1}{2_1},{2_1}{2_1}{2_1} |
{1_2}{3_3},{2_2}} |
{1_3}{1_3},{2_3}}
```

2G.21

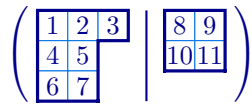
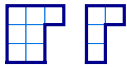
separators (default: **true**)

[accepts: true/false]

no separators (default: **false**)

[accepts: false/true]

When **separators** is true, delimiters are drawn around the component diagrams in a multitableaux and a multidiagram and separators are drawn between the component shapes, and **no separators** is the inverse option. By default, separators and delimiters are drawn.



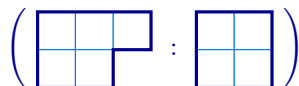
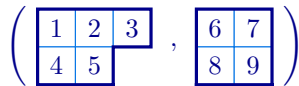
```
\Multidiagram[no separators,scale=0.5]
{3,2^2|2,1^2}
\Multitableau[separators,scale=0.8]
{123,45,67 | 89,{10}{11}}
```

2G.22

separator (default: |)

[accepts: character]

The **separator** determines what is printed between the component diagrams in multitableaux and multi-partition. By default, **separator**=— draws a straight line between the components, but any character can be used.



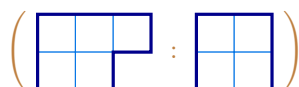
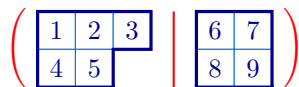
```
\Multitableau[separator={,}] { 123,45 | 67,89}
\Multidiagram[separator=:] { 3,2 | 2^2 }
```

2G.23

separator colour (default:)

[accepts: L^AT_EX colour]

The **separator colour** sets the colour of the delimiters and the separator between component diagrams. By default, the separators are the same colour as the diagram border.



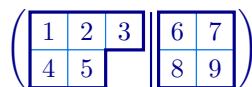
```
\Multitableau[separator colour=red] {123,45 | 67,89}
\Multidiagram[separator color=brown, separator=:]
{3,2 | 2^2 }
```

2G.24

separation (default: 0.3)

[accepts: a decimal number (the distance in cm)]

The **separation** key sets the distance between the separators and the component tableaux and diagrams.



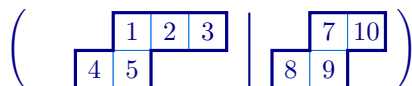
```
\Multitableau[separation=0.1] {123,45 | 67,89}
\Multidiagram[separation=0.8] {3,2 | 2^2 }
```

2G.25

skew

a |-separated list of partitions

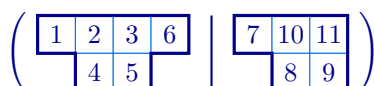
Skew multishapes can be drawn using the **skew** option to specify a |-separated list of inner shapes.



```
\Multitableau[skew={2,1|1}] { 123,45 | 7{10},89}
\Multidiagram[skew={1|2,1}, skew boxes] { 3,2 | 2^2 }
```

2G.26

There is no command for drawing shifted multitableau or multidiagrams, however, such diagrams can be drawn using the **skew** key. In the same way, it is possible to draw multitableaux and multidiagrams that have components which are a mixture of non-skew, skew and shifted shapes.

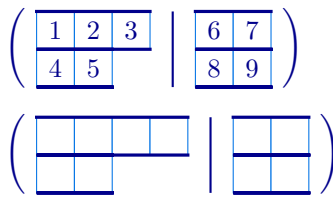


```
\Multitableau[skew={0,1|0,1}] { 1236,45 | 7{10}{11},89}
\Multidiagram[skew={|0,1}] { 3,2 | 2,1 }
```

2G.27

tabloid

Use the `tabloid` key for multitableid diagrams and tableaux.



```
\Multitableau[tabloid]{ 123,45 | 67,89}
\Multidiagram[tabloid]{ 4,2 | 2^2 }
```

2G.28

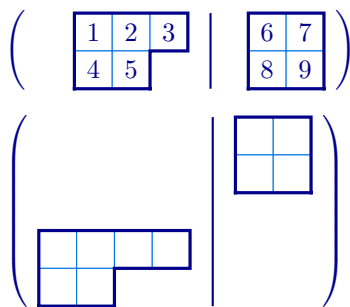
`xoffsets` (default: 0)

[accepts: comma separated list of x -offsets]

`yoffsets` (default: 0)

[accepts: comma separated list of y -offsets]

The `\Multitableau` and `\Multidiagram` commands try to draw the diagrams as compactly as possible with their first rows aligned, subject to the value of `separation`. Use the `xoffsets` to offset the x -coordinates of the component diagrams and `yoffsets` to offset their y -coordinates.



```
\Multitableau[xoffsets={0.5,0.2}]{ 123,45 | 67,89}
\Multidiagram[yoffsets={0,1.5}]{ 4,2 | 2^2 }
```

2G.29

When using `xoffsets` or `yoffsets`, you may want to use `no separators` to disable the separators between component shapes.

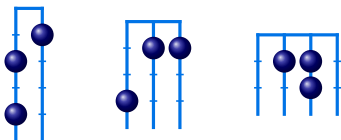
3. ABACUSES

Gordon James introduced the abacus, with beads and runners, as another way to represent partition. The abacus used particularly in modular representation theory, where the runner of runners is (quantum) characteristic.

The partition corresponding to an abacus is given by counting the number of empty bead positions before each bead on the abacus. The `\Abacus` command, which draws an abacus representation of a partition, has a similar syntax to the tableau and diagram commands.

```
\Abacus (x,y) [options] {number of runners} {bead specifications}
```

As with other `aTableau` commands, the (x,y) -coordinates are necessary if, and only if, the abacus is part of a `tikzpicture` environment. The number of runners is a positive integer that specifies the number of abacus runners. In their simplest form, the bead specifications are a comma separated list of non-negative integers specifying a partition, with exponential notation and optional styling and bead labels being supported.

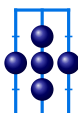


```
\Abacus{2}{4,1,1}
\Abacus{3}{4,1,1}
\Abacus{4}{4,1,1}
```

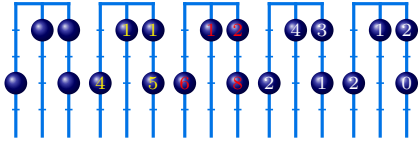
3.1

3A. Bead specifications.

3B. Abacus options.

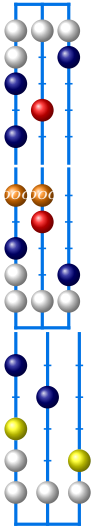


```
\Abacus[top=frame]{3}{3,2^3,1}
```



```
\Abacus{3}{5,4,1,1}
\Abacus[entries=shape,bead text=yellow]{3}{5,4,1,1}
\Abacus[entries=beta,bead text=red]{3}{5,4,1,1}
\Abacus[entries=rows]{3}{5,4,1,1}
\Abacus[entries=residue]{3}{5,4,1,1}
```

3B.2



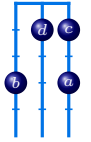
```
\tikzset{R/.style={ball color=red},
W/.style={ball color=white},
O/.style={ball color=orange}
}
```

3B.3

```
\Abacus{3}{5,[R]4,1^2,[W]0^4}

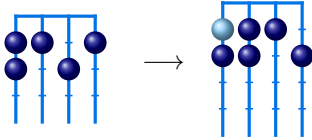
\Abacus[north]{3}{[O]5^2_{ooo},[R]4,1^2,[W]0^4}

\Abacus[north,scale=1.2,star style={ball
color=yellow}]{3}{5,4,*1^2,[W]0^4}
```



```
\Abacus[bead text=white,bead
font=\tiny]{3}{5_a,4_b,1_c,1_d}
```

3B.4



```
\tikzset{B/.style={ball color=LightSkyBlue}}
\Abacus[beta numbers]{4}{6,4,3,1,0}
$\longrightarrow$
\Abacus[beta numbers]{4}{7,5,4,2,1,[B]0}
```

3B.5

3C. **Colours.** The **aTableau** package defines and uses the following colours:

Colour	Name	HTML
	aTableauMain	00008B
	aTableauInner	0073E6
	aTableauSkew	818589
	aTableauSkewFill	F8F8F8
	aTableauStarStyle	E6F7FF

All of these colours are defined as HTML colours using the `\definecolor` command provided by **xcolor**.

It is possible to change the colours used in the diagrams created by **aTableau** by changing these colours. However, we recommend using the key-value interface instead.

4. FEATURE REQUESTS AND BUG REPORTS

Please make any feature requests and bug reports on the package github page

github.com/AndrewMathas/aTableau/.

Please give as much detail as possible when making such requests.

Bug reports must be accompanied by a *minimal working example*, which is the smallest possible amount of $\text{\LaTeX}$ code that compiles and demonstrates the problem. This is necessary because if I cannot replicate your problem, then it is unlikely that I will be able to fix it.

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